

True or false?

$(9 + 4) \times 2$ is 2 more than $9 + 4$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q3/5

1601

QUESTION ID

S43-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q3/5

Which expression is 4 times as large as $125 - 25$?

- A. $(125 - 25) \div 4$
- B. $4 \times (125 - 25)$
- C. $125 - 25 \times 4$
- D. $4 + (125 - 25)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q4/5

1602

QUESTION ID

S43-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $4 \times (125 - 25)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q4/5

Without calculating, how many times as large is
 $10 \times (63 + 7)$ as $63 + 7$?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q5/5

1603

QUESTION ID

S43-B3-Q5

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • 6.EE.A.2.b

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY

Reduced copy of the question page, with its answer

Simplify.

$$4x + 3 - 2x + 5$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q1/5

1605

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

 $2x + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q1/5

1605

ANSWER KEY

Reduced copy of the question page, with its answer

Simplify.

$$0.5y + 1.25y$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q2/5

1606

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.75y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q2/5

1606

Which pair shows like terms?

- A.** $3a$ and $-7a$
- B.** x^2 and x
- C.** $3a$ and $3b$
- D.** $4x$ and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q3/5

1607

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $3a$ and $-7a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q3/5

Simplify.

$$-3(x - 2) + 5x$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q4/5

1608

QUESTION ID

S43-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$2x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2}x + \frac{1}{4}x = \frac{3}{4}x$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q5/5

1609

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 6.EE.A.2.b

FORWARD 2

Interpret the parts of a linear function • Grade 8 • Functions • 8.F.B.4

The cost to bowl is $C = 4.5n + 12$, where n is the number of games.

What does 12 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q1/5

1611

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The fixed \$12 cost that is paid no matter how many games are bowled (for example, a shoe rental or entry fee)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q1/5

In $y = -3x + 7$, what is the rate of change?

- A. x
- B. 3
- C. -3
- D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q2/5

1612

QUESTION ID

S43-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q2/5

The gas left in a car's tank is $G = 14 - 0.04m$, where m is miles driven.

What is the initial value?

Initial value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q3/5

1613

QUESTION ID

S43-F2-Q3

TYPE

Short answer

CORRECT ANSWER

14 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q3/5

A car's distance from home is $d = 60t + 25$, where t is hours.

What does 60 represent?

- A.** The speed in miles per hour **B.** The starting distance
C. The total distance **D.** The total time

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q4/5

1614

QUESTION ID

S43-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The speed in miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q4/5

True or false?

In $y = 8x$, the initial value is 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q5/5

1615

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q5/5

Evaluate expressions and formulas for given values

MAIN	Grade 6 main question	6.EE.A.2.c
BACKWARD 1	Whole-number exponents	6.EE.A.1
BACKWARD 2	Order of operations with grouping symbols	5.OA.A.1
BACKWARD 3	A letter stands for a number	3.OA.D.8
BACKWARD 4	Multiplying fractions	5.NF.B.4.a
FORWARD 1	Evaluate expressions with rational numbers	7.NS.A.3
FORWARD 2	Volume formulas for cylinders, cones, and spheres	8.G.C.9

SET 44 • 6.EE.A.2.c

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x^2 + 2$ when $x = 4$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q1/5

1618

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q1/5

The volume of a cube is $V = s^3$.

Find V when $s = \frac{2}{3}$ foot.

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q2/5

1619

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

8/27 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q2/5

The perimeter of a rectangle is $P = 2l + 2w$.

Find P when $l = 7.5$ cm and $w = 4$ cm.

$P =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q3/5

1620

QUESTION ID

S44-M-Q3

TYPE

Short answer

CORRECT ANSWER

23 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q3/5

Evaluate $2(a + b)$ when $a = 3.5$ and $b = 4$.

- A.** 7.5
- B.** 9.5
- C.** 11.5
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q4/5

1621

QUESTION ID

S44-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q4/5

Evaluate $\frac{x}{4} + 5y$ when $x = 12$ and $y = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q5/5

1622

QUESTION ID

S44-M-Q5

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q5/5

SET 44 • 6.EE.A.2.c

BACKWARD 1

Whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$4^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

1624

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

Evaluate.

$$\left(\frac{1}{2}\right)^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

1625

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

True or false?

$$3^2 = 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

1626

QUESTION ID

S44-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

$$2^5 = ?$$

- A.** 10
- B.** 16
- C.** 25
- D.** 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

1627

QUESTION ID

S44-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

Evaluate.

$$5^2 + 1$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

1628

QUESTION ID

S44-B1-Q5

TYPE

Short answer

CORRECT ANSWER

26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 6.EE.A.2.c

BACKWARD 2

Order of operations with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$3 \times (16 + 2)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q1/5

1630

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q1/5

Evaluate.

$$2 \times (6 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q2/5

1631

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q2/5

True or false?

$$(3 + 4) \times 2 = 11$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q3/5

1632

QUESTION ID

S44-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q3/5

Evaluate.

$$(12 \div 4) + 10$$

- A. 1
- B. 6
- C. 13
- D. 22

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q4/5

1633

QUESTION ID

S44-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q4/5

Evaluate.

$$[2 \times (5 + 3)] - 4$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q5/5

1634

QUESTION ID

S44-B2-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 6.EE.A.2.c

BACKWARD 3

A letter stands for a number • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

If $n = 6$, what is $n + 9$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q1/5

1636

QUESTION ID

S44-B3-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q1/5

True or false?

If $a = 4$, then $a \times 5 = 20$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q2/5

1637

QUESTION ID

S44-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q2/5

If $k = 7$, what is $3 \times k$?

- A.** 10
- B.** 21
- C.** 37
- D.** 73

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q3/5

1638

QUESTION ID

S44-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q3/5

If $m = 20$, what is $m - 8$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q4/5

1639

QUESTION ID

S44-B3-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q4/5

If $t = 3$, what is $4 \times t + 2$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q5/5

1640

QUESTION ID

S44-B3-Q5

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • 6.EE.A.2.c

BACKWARD 4

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q1/5

1642

QUESTION ID

S44-B4-Q1

TYPE

Short answer

CORRECT ANSWER

8/27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q1/5

Multiply.

$$6 \times \frac{1}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q2/5

1643

QUESTION ID

S44-B4-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q2/5

True or false?

$$\frac{2}{3} \times \frac{2}{3} = \frac{4}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q3/5

1644

QUESTION ID

S44-B4-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q3/5

$$\frac{3}{4} \times 8 = ?$$

- A. $\frac{3}{32}$
- B. 2
- C. 6
- D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q4/5

1645

QUESTION ID

S44-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q4/5

Multiply.

$$\frac{1}{2} \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q5/5

1646

QUESTION ID

S44-B4-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q5/5

SET 44 • 6.EE.A.2.c

FORWARD 1

Evaluate expressions with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Use $C = \frac{5}{9}(F - 32)$ to find C when $F = -4$.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q1/5

1648

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q1/5

Evaluate $-2x + 5$ when $x = -3.5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q2/5

1649

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q2/5

Evaluate $a^2 - b$ when $a = -3$ and $b = 4$.

- A.** -13
- B.** -5
- C.** 5
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q3/5

1650

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q3/5

Evaluate $\frac{(x+y)}{2}$ when $x = -6.5$ and $y = 2.5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q4/5

1651

QUESTION ID

S44-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q4/5

True or false?

When $x = -2$, the value of $x^3 + 1$ is -7 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q5/5

1652

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 6.EE.A.2.c

FORWARD 2

Volume formulas for cylinders, cones, and spheres • Grade 8 • Geometry •
8.G.C.9

A cylinder has a radius of 3 cm and a height of 10 cm.

Find its volume. Use 3.14 for π .

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q1/5

1654

GRADING NOTE 90π

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

282.6 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

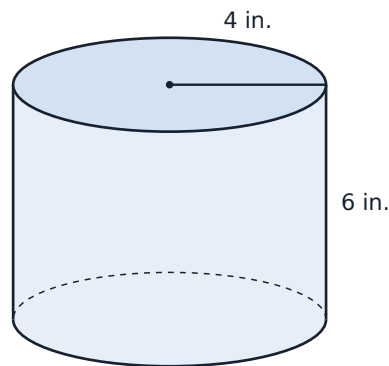
ID S44-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q1/5

Find the volume of the cylinder.

Use 3.14 for π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q2/5

1655

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

301.44 cubic inches

GRADING NOTE 96π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q2/5

A cone has a radius of 3 units and a height of 4 units.

What is its volume in terms of π ?

- A. 36π
- B. 48π
- C. 4π
- D. 12π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q3/5

1656

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12π

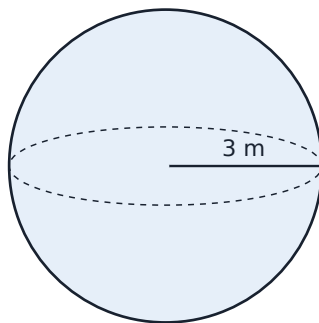
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q3/5

Find the volume of the sphere
in terms of π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q4/5

1657

QUESTION ID

S44-F2-Q4

TYPE

Short answer

CORRECT ANSWER

36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 20π .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q5/5

1658

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q5/5

Use the distributive property to expand expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Distributive property with numbers	3.OA.B.5
BACKWARD 2	Equal groups	3.OA.A.1
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Expand linear expressions with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by expanding and collecting like terms	8.EE.C.7.b

SET 45 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(4 + x)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q1/5

1661

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

12 + 3x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q1/5

Expand.

$$5(3a - 2)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q2/5

1662

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

15a – 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q2/5

Which expression is equivalent to $7(2m + 5)$?

- A.** $9m + 35$
- B.** $2m + 35$
- C.** $14m + 5$
- D.** $14m + 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q3/5

1663

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $14m + 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q3/5

Expand.

$$2(x + 3y + 4)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q4/5

1664

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

$$2x + 6y + 8$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q4/5

Expand.

$$4(0.5y + 3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q5/5

1665

QUESTION ID

S45-M-Q5

TYPE

Short answer

CORRECT ANSWER

$2y + 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q5/5

SET 45 • 6.EE.A.3

BACKWARD 1

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$6 \times (4 + 3) = 6 \times 4 + 6 \times 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q1/5

1667

QUESTION ID

S45-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q1/5

$$5 \times (2 + 7) = 5 \times 2 + 5 \times \underline{\quad}$$

Blank: _____

ID S45-B1-Q2

1668

7

ID S45-B1-Q2

1668

Which is equal to 3×14 ?

- A. $30 + 4 + 3$
- B. $3 \times 10 + 4$
- C. $3 \times 10 + 3 \times 4$
- D. $3 + 10 \times 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q3/5

1669

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $3 \times 10 + 3 \times 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q3/5

True or false?

$$7 \times 12 = 7 \times 10 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q4/5

1670

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q4/5

Blank: _____

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q5/5

1671

8 × 1

1671

SET 45 • 6.EE.A.3

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 2)$ means 3 groups of $(4 + 2)$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q1/5

1673

QUESTION ID

S45-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q1/5

Which means 4 groups of $(5 + 1)$?

- A. $4 + (5 + 1)$
- B. $4 \times (5 + 1)$
- C. $4 \times 5 + 1$
- D. $(4 + 5) \times 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q2/5

1674

QUESTION ID

S45-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $4 \times (5 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q2/5

How many groups of $(2 + 6)$ are in $5 \times (2 + 6)$?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q3/5

1675

QUESTION ID

S45-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q3/5

True or false?

$$2 \times (3 + 4) = (3 + 4) + (3 + 4)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q4/5

1676

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q4/5

Write 6×3 as a sum of equal groups.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q5/5

1677

QUESTION ID

S45-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6 + 6 + 6

GRADING NOTE $3 + 3 + 3 + 3 + 3 + 3$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 6.EE.A.3

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4 \times 0.5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q1/5

1679

QUESTION ID

S45-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q1/5

1679

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3 \times 1.2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q2/5

1680

QUESTION ID

S45-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q2/5

1680

True or false?

$$6 \times 0.25 = 1.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q3/5

1681

QUESTION ID

S45-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q3/5

$$5 \times 0.4 = ?$$

- A.** 0.2
- B.** 2
- C.** 5.4
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q4/5

1682

QUESTION ID

S45-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$8 \times 0.75$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q5/5

1683

QUESTION ID

S45-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q5/5

1683

SET 45 • 6.EE.A.3

FORWARD 1

Expand linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$$\frac{1}{2}(8x - 6)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q1/5

1685

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$4x - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q1/5

Expand.

$$-4(2y - 3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q2/5

1686

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$-8y + 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q2/5

Which is equivalent to $0.5(6a + 10)$?

- A. $3a + 5$
- B. $6a + 5$
- C. $3.5a + 5$
- D. $3a + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q3/5

1687

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $3a + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q3/5

Expand.

$$-\frac{2}{3}(9m + 3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q4/5

1688

QUESTION ID

S45-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$-6m - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q4/5

True or false?

$$-2(x - 7) = -2x + 14$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q5/5

1689

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 6.EE.A.3

FORWARD 2

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$3(x + 2) = 5x - 4$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q1/5

1691

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q1/5

Solve for x.

$$2(4x - 1) - 3x = 18$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q2/5

1692

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q2/5

Solve for x.

$$-2(x - 5) = 3(x + 5)$$

- A.** -5
- B.** -1
- C.** 1
- D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q3/5

1693

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q3/5

Solve for x.

$$6(x - 1) = 4(x + 2)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q4/5

1694

QUESTION ID

S45-F2-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Solve for x.

$$5(2x + 1) = 3(3x + 4)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q5/5

1695

QUESTION ID

S45-F2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q5/5

1695

Use the distributive property to factor expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Greatest common factor	6.NS.B.4
BACKWARD 2	Distributive property with numbers	3.OA.B.5
BACKWARD 3	Factors	4.OA.B.4
FORWARD 1	Factor linear expressions with rational coefficients	7.EE.A.1
FORWARD 2	Use factoring to decide how many solutions an equation has	8.EE.C.7.a

SET 46 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Factor using the greatest common factor.

$$24x + 16$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q1/5

1698

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

8(3x + 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q1/5

Factor using the greatest common factor.

$$18y + 30$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q2/5

1699

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

$6(3y + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q2/5

Which shows $15a + 25b$ factored using the greatest common factor?

- A. $15(a + 10b)$
- B. $5(3a + 5b)$
- C. $3(5a + 25b)$
- D. $5(3a + 25b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q3/5

1700

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $5(3a + 5b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q3/5

Factor using the greatest common factor.

$$36m + 27n$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q4/5

1701

QUESTION ID

S46-M-Q4

TYPE

Short answer

CORRECT ANSWER

9(4m + 3n)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q4/5

Factor using the greatest common factor.

$$14x + 21$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q5/5

1702

QUESTION ID

S46-M-Q5

TYPE

Short answer

CORRECT ANSWER

$7(2x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q5/5

SET 46 • 6.EE.A.3

BACKWARD 1

Greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 16.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q1/5

1704

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q1/5

Find the greatest common factor of 18 and 30.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q2/5

1705

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q2/5

True or false?

The greatest common factor of 15 and 25 is 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q3/5

1706

QUESTION ID

S46-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q3/5

What is the greatest common factor of 36 and 27?

- A.** 3
- B.** 6
- C.** 9
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q4/5

1707

QUESTION ID

S46-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q4/5

Find the greatest common factor of 14 and 21.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q5/5

1708

QUESTION ID

S46-B1-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 6.EE.A.3

BACKWARD 2

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 \times 3 + 8 \times 2 = 8 \times (3 + 2)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q1/5

1710

QUESTION ID

S46-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q1/5

Fill in the blank.

$$6 \times 4 + 6 \times 5 = 6 \times (4 + \underline{\quad})$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q2/5

1711

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q2/5

Which is equal to $7 \times 2 + 7 \times 9$?

- A. $14 + 9$
- B. $7 \times 2 \times 9$
- C. $7 \times (2 + 9)$
- D. $7 + 2 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q3/5

1712

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $7 \times (2 + 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q3/5

True or false?

$$5 \times 6 + 3 \times 6 = 6 \times (5 + 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q4/5

1713

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q4/5

Fill in the blank.

$$9 \times 7 + 9 \times 3 = \underline{\hspace{1cm}} \times (7 + 3)$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q5/5

1714

QUESTION ID

S46-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 6.EE.A.3

BACKWARD 3

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 16.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q1/5

1716

QUESTION ID

S46-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, 4, 8, 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q1/5

True or false?

6 is a factor of 30.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q2/5

1717

QUESTION ID

S46-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q2/5

Which number is a factor of 27?

- A.** 4
- B.** 6
- C.** 9
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q3/5

1718

QUESTION ID

S46-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q3/5

List all the factors of 21.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q4/5

1719

QUESTION ID

S46-B3-Q4

TYPE

Short answer

CORRECT ANSWER

1, 3, 7, 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q4/5

True or false?

8 is a factor of 36.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q5/5

1720

QUESTION ID

S46-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • 6.EE.A.3

FORWARD 1

Factor linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor out 2.5.

$$2.5x + 10$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q1/5

1722

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5(x + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q1/5

Factor out -4 .

$$-4x + 12$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q2/5

1723

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$-4(x - 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q2/5

Which is equivalent to $\frac{1}{2}x + 3$?

A. $\frac{1}{2}(x + \frac{3}{2})$

B. $\frac{1}{2}(x + 3)$

C. $\frac{1}{2}(x + 6)$

D. $2(x + 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q3/5

1724

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(\frac{1}{2})(x + 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Factor out 0.6.

$$0.6y - 1.8$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q4/5

1725

QUESTION ID

S46-F1-Q4

TYPE

Short answer

CORRECT ANSWER

 $0.6(y - 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q4/5

1725

True or false?

$$-5a - 15 = -5(a + 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q5/5

1726

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 6.EE.A.3

FORWARD 2

Use factoring to decide how many solutions an equation has • Grade 8 •
Expressions & Equations • 8.EE.C.7.a

Factor the left side to decide. How many solutions does the equation have?

$$6x + 9 = 3(2x + 3)$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q1/5

1728

QUESTION ID

S46-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$12x + 8 = 4(3x + 1)$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q2/5

1729

QUESTION ID

S46-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q2/5

How many solutions does the equation have?

$$10x - 15 = 5(2x - 3)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q3/5

1730

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q3/5

How many solutions does the equation have?

$$8x + 4 = 4(x + 1)$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q4/5

1731

QUESTION ID

S46-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q4/5

What number goes in the box so that the equation has infinitely many solutions?

$$14x + 21 = 7(2x + \square)$$

$\square =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q5/5

1732

QUESTION ID

S46-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q5/5

Combine like terms to write equivalent expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Repeated addition as multiplication	3.OA.A.1
BACKWARD 2	Changing the order and grouping of addends	3.OA.B.5
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by collecting like terms	8.EE.C.7.b

SET 47 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Write an equivalent expression with one term.

$$y + y + y + y$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q1/5

1735

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

4y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q1/5

Simplify.

$$5x + 3x$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q2/5

1736

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

8x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q2/5

Which expression is equivalent to $7a + 2 + 3a$?

- A. $7a + 5a$
- B. $10a + 5$
- C. $10a + 2$
- D. $12a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q3/5

1737

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $10a + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q3/5

Simplify.

$$4m + 6 + 2m + 1$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q4/5

1738

QUESTION ID

S47-M-Q4

TYPE

Short answer

CORRECT ANSWER

6m + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q4/5

Simplify.

$$9n - 4n + 3$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q5/5

1739

QUESTION ID

S47-M-Q5

TYPE

Short answer

CORRECT ANSWER

$5n + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q5/5

SET 47 • 6.EE.A.3

BACKWARD 1

Repeated addition as multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$4 + 4 + 4 = 3 \times 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q1/5

1741

QUESTION ID

S47-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q1/5

Write $7 + 7 + 7 + 7$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q2/5

1742

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4 × 7

GRADING NOTE 7×4 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q2/5

What does 4×9 mean?

- A.** 4 groups of 9
- B.** 9 minus 4
- C.** 4 more than 9
- D.** The number 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q3/5

1743

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4 groups of 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q3/5

True or false?

$$2 + 2 + 2 + 2 + 2 = 2 \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q4/5

1744

QUESTION ID

S47-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q4/5

Write $9 + 9$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q5/5

1745

QUESTION ID

S47-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2 × 9

GRADING NOTE 9×2 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 6.EE.A.3

BACKWARD 2

Changing the order and grouping of addends • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$4 + 6 + 2 = 4 + 2 + 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q1/5

1747

QUESTION ID

S47-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q1/5

Fill in the blank.

$$7 + 5 + 3 = 7 + 3 + \underline{\quad}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q2/5

1748

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q2/5

Which is equal to $(8 + 2) + 5$?

- A. $8 + 2 \times 5$
- B. $8 \times (2 + 5)$
- C. $(8 - 2) + 5$
- D. $8 + (2 + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q3/5

1749

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $8 + (2 + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q3/5

True or false?

$$9 + 1 + 9 = 9 + 9 + 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q4/5

1750

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q4/5

$$6 + 3 + 4 = 6 + 4 + \underline{\quad}$$

Blank: _____

ID S47-B2-Q5

1751

S47-B2-Q5

Short answer

3

ID S47-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 6.EE.A.3

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$$2.25y + 0.75y$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q1/5

1753

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q1/5

Simplify.

$$\frac{1}{2}x + \frac{1}{4}x$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q2/5

1754

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(\frac{3}{4})x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q2/5

Simplify.

$$6x - 2 + 3x + 7$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q3/5

1755

QUESTION ID

S47-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$9x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q3/5

Simplify.

$$-3a + 7a - 2$$

- A. $4a - 2$
- B. $4a + 2$
- C. $2a$
- D. $-10a - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q4/5

1756

QUESTION ID

S47-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $4a - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q4/5

True or false?

$$2.5\text{m} - \text{m} = 1.5\text{m}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q5/5

1757

QUESTION ID

S47-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 6.EE.A.3

FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$3x + 5x - 4 = 20$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q1/5

1759

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q1/5

Solve for y.

$$7y - 2y + 3 = 2y + 18$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q2/5

1760

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q2/5

Solve for a.

$$4a + 6 - a = 2a + 10$$

A. -4

B. $\frac{16}{5}$

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q3/5

1761

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q3/5

Solve for n.

$$0.5n + 1.5n - 3 = n + 5$$

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q4/5

1762

QUESTION ID

S47-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q4/5

Solve for k.

$$6k - 2k + 1 = 3k - 4$$

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q5/5

1763

QUESTION ID

S47-F2-Q5

TYPE

Short answer

CORRECT ANSWER

-5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q5/5

Identify when two expressions are equivalent

MAIN	Grade 6 main question	6.EE.A.4
BACKWARD 1	Commutative and associative properties	3.OA.B.5
BACKWARD 2	Evaluate an expression for a given value	6.EE.A.2.c
BACKWARD 3	Equal groups	3.OA.A.1
FORWARD 1	Rewrite expressions to show how quantities are related	7.EE.A.2
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 48 • 6.EE.A.4

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.4

True or false?

$m + m + m + m$ and $4m$ are equivalent expressions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q1/5

1766

QUESTION ID

S48-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q1/5

Which expression is equivalent to $2(x + 4)$?

- A. $x + 8$
- B. $2x + 4$
- C. $2x + 6$
- D. $2x + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q2/5

1767

QUESTION ID

S48-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $2x + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q2/5

True or false?

$3(x + 2)$ and $3x + 2$ are equivalent expressions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q3/5

1768

QUESTION ID

S48-M-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q3/5

Which expression is equivalent to $6a + 4a$?

- A. $6a + 4$
- B. $10a$
- C. $24a$
- D. $10a^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q4/5

1769

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $10a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q4/5

Which pair of expressions is equivalent?

- A. $5(a + 3)$ and $5a + 15$
- B. $5(a + 3)$ and $a + 15$
- C. $5(a + 3)$ and $5a + 3$
- D. $5(a + 3)$ and $8a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q5/5

1770

QUESTION ID

S48-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

**A. $5(a + 3)$ and
 $5a + 15$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q5/5

SET 48 • 6.EE.A.4

BACKWARD 1

Commutative and associative properties • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 + 5 = 5 + 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q1/5

1772

QUESTION ID

S48-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q1/5

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q2/5

1773

QUESTION ID

S48-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q2/5

Which equation shows the commutative property of multiplication?

- A. $6 \times 7 = 42$
- B. $6 \times 7 = 7 \times 6$
- C. $6 \times (7 + 1) = 6 \times 7 + 6$
- D. $6 + 0 = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q3/5

1774

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $6 \times 7 = 7 \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q3/5

True or false?

$$9 - 4 = 4 - 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q4/5

1775

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q4/5

Fill in the blank.

$$3 \times 8 = 8 \times \underline{\quad}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q5/5

1776

QUESTION ID

S48-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 6.EE.A.4

BACKWARD 2

Evaluate an expression for a given value • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4x + 2x$ when $x = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q1/5

1778

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q1/5

Evaluate $3(x + 2)$ when $x = 5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q2/5

1779

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q2/5

True or false?

When $x = 2$, the value of $3x + 2$ is 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q3/5

1780

QUESTION ID

S48-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q3/5

Evaluate $2(x + 4)$ when $x = 1$.

- A.** 6
- B.** 9
- C.** 10
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q4/5

1781

QUESTION ID

S48-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q4/5

Evaluate $5a + 15$ when $a = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q5/5

1782

QUESTION ID

S48-B2-Q5

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 6.EE.A.4

BACKWARD 3

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$6 + 6 + 6 + 6 = 4 \times 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q1/5

1784

QUESTION ID

S48-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q1/5

Which expression means 3 groups of 8?

- A. $3 + 8$
- B. 38
- C. $8 - 3$
- D. 3×8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q2/5

1785

QUESTION ID

S48-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 3×8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q2/5

Write $5 + 5 + 5$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q3/5

1786

QUESTION ID

S48-B3-Q3

TYPE

Short answer

CORRECT ANSWER

5 × 3

GRADING NOTE 3×5 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q3/5

True or false?

$$7 \times 2 = 7 + 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q4/5

1787

QUESTION ID

S48-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q4/5

How many groups of 4 are in $4 + 4 + 4 + 4 + 4$?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q5/5

1788

QUESTION ID

S48-B3-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • 6.EE.A.4

FORWARD 1

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price c plus an 8% tax can be written as $c + 0.08c$.

Which expression is equivalent?

- A. $0.08c$
- B. $1.08c$
- C. $1.8c$
- D. $8.08c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q1/5

1790

QUESTION ID

S48-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $1.08c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q1/5

True or false?

$$a - 0.2a = 0.8a$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q2/5

1791

QUESTION ID

S48-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q2/5

Write an expression with one term that is equivalent to $n - 0.25n$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q3/5

1792

QUESTION ID

S48-F1-Q3

TYPE

Short answer

CORRECT ANSWER

0.75n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q3/5

Which expression shows the price p after a 30% discount?

- A. $0.3p$
- B. $0.7p$
- C. $1.3p$
- D. $p - 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q4/5

1793

QUESTION ID

S48-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $0.7p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q4/5

The perimeter of a square is $s + s + s + s$.

Write an equivalent expression with one term.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q5/5

1794

QUESTION ID

S48-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 6.EE.A.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

True or false?

The equation $2(x + 3) = 2x + 6$ has infinitely many solutions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q1/5

1796

QUESTION ID

S48-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$5x - 3 = 5x + 1$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q2/5

1797

QUESTION ID

S48-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q2/5

How many solutions does the equation have?

$$3(2x + 4) = 6x + 12$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q3/5

1798

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q3/5

Write a number in the box so the equation has no solution.

$$4x + 7 = 4x + \square$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q4/5

1799

QUESTION ID

S48-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**Any number except
7**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q4/5

Which equation has exactly one solution?

- A. $3x + 1 = 2x + 1$
- B. $3x + 1 = 3x + 2$
- C. $3x + 1 = 3x + 1$
- D. $3(x + 1) = 3x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q5/5

1800

QUESTION ID

S48-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $3x + 1 = 2x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q5/5

Use substitution to decide whether a number makes an equation or inequality true

MAIN	Grade 6 main question	6.EE.B.5
BACKWARD 1	Evaluate an expression	6.EE.A.2.c
BACKWARD 2	Comparing numbers	2.NBT.A.4
BACKWARD 3	Meaning of the equal sign	1.OA.D.7
FORWARD 1	Solve equations of the form $px + q = r$ and $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solutions of a system of two linear equations	8.EE.C.8.a

SET 49 • 6.EE.B.5

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.5

Which value from the set $\{2, 4, 6, 8\}$ makes $3x + 1 = 19$ true?

- A. 2
- B. 4
- C. 6
- D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q1/5

1803

QUESTION ID

S49-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q1/5

True or false?

$x = 5$ is a solution of $2x - 3 > 8$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q2/5

1804

QUESTION ID

S49-M-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q2/5

Which numbers from the set $\{1, 3, 5, 7\}$ make $x + 4 < 10$ true?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q3/5

1805

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

1, 3, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q3/5

Is $y = 2.5$ a solution of $4y = 10$?

A. Yes

B. No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q4/5

1806

QUESTION ID

S49-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q4/5

Which value from the set $\{12, 15, 18\}$ makes $n \div 3 = 6$ true?

$n =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q5/5

1807

QUESTION ID

S49-M-Q5

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q5/5

SET 49 • 6.EE.B.5

BACKWARD 1

Evaluate an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x + 1$ when $x = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q1/5

1809

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q1/5

Evaluate $2x - 3$ when $x = 5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q2/5

1810

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q2/5

True or false?

When $y = 4$, the value of $5y$ is 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q3/5

1811

QUESTION ID

S49-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q3/5

Evaluate $x + 4$ when $x = 7$.

- A. 3
- B. 11
- C. 28
- D. 47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q4/5

1812

QUESTION ID

S49-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q4/5

Evaluate $n \div 3$ when $n = 15$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q5/5

1813

QUESTION ID

S49-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 6.EE.B.5

BACKWARD 2

Comparing numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

19 ____ 21

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q1/5

1815

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q1/5

True or false?

$$7 > 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q2/5

1816

QUESTION ID

S49-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q2/5

Which statement is true?

- A.** $9 > 6$
- B.** $10 < 5$
- C.** $11 < 10$
- D.** $6 > 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q3/5

1817

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $9 > 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q3/5

True or false?

$$24 = 24$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q4/5

1818

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $<$, or $=$ to compare.

105 ____ 150

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q5/5

1819

QUESTION ID

S49-B2-Q5

TYPE

Short answer

CORRECT ANSWER

 $<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q5/5

1819

SET 49 • 6.EE.B.5

BACKWARD 3

Meaning of the equal sign • Grade 1 • Operations & Algebraic Thinking •
1.OA.D.7

True or false?

$$4 + 5 = 9 + 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q1/5

1821

QUESTION ID

S49-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q1/5

True or false?

$$12 = 15 - 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q2/5

1822

QUESTION ID

S49-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q2/5

True or false?

$$3 + 6 = 5 + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q3/5

1823

QUESTION ID

S49-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q3/5

Which equation is true?

A. $4 + 3 = 5 + 3$

B. $4 + 3 = 5 + 2$

C. $7 = 4 - 3$

D. $4 + 3 = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q4/5

1824

QUESTION ID

S49-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $4 + 3 = 5 + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q4/5

True or false?

$$10 - 2 = 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q5/5

1825

QUESTION ID

S49-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • 6.EE.B.5

FORWARD 1

Solve equations of the form $px + q = r$ and $p(x + q) = r$ • Grade 7 •
Expressions & Equations • 7.EE.B.4.a

ANSWER KEY

Reduced copy of the question page, with its answer

Solve for x.

$$3x + 7 = 22$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q1/5

1827

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q1/5

1827

Solve for x.

$$4(x - 2) = 20$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q2/5

1828

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q2/5

Solve for x.

$$0.5x + 6 = 10$$

- A.** 2
- B.** 8
- C.** 20
- D.** 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q3/5

1829

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q3/5

Solve for x.

$$\frac{2}{3}x - 4 = 10$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q4/5

1830

QUESTION ID

S49-F1-Q4

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q4/5

A movie ticket costs \$12. Popcorn costs p dollars. Two tickets and one popcorn cost \$31.

What is p ?

$p =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q5/5

1831

QUESTION ID

S49-F1-Q5

TYPE

Short answer

CORRECT ANSWER

\$7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q5/5

SET 49 • 6.EE.B.5

FORWARD 2

Solutions of a system of two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.a

True or false?

$(2, 3)$ is a solution of both $y = x + 1$ and $y = 2x - 1$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q1/5

1833

QUESTION ID

S49-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q1/5

Which point is a solution of both $y = 2x$ and $y = x + 3$?

- A. (0, 3)
- B. (2, 5)
- C. (1, 2)
- D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q2/5

1834

QUESTION ID

S49-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. (3, 6)

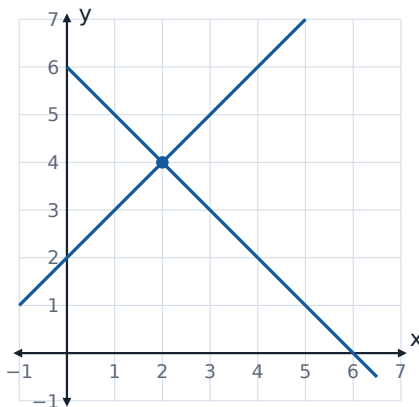
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q2/5

What is the solution of the system shown by the graph?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q3/5

1835

QUESTION ID

S49-F2-Q3

TYPE

Short answer

CORRECT ANSWER

(2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q3/5

True or false?

(1, 4) is a solution of both $y = 3x + 1$ and $y = -x + 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q4/5

1836

QUESTION ID

S49-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q4/5

Is $(0, 2)$ a solution of both $y = 2x + 2$ and $y = x - 2$?

A. Yes

B. No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q5/5

1837

QUESTION ID

S49-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q5/5

Use variables to write expressions for real-world problems

MAIN	Grade 6 main question	6.EE.B.6
BACKWARD 1	Word problems with an unknown number	3.OA.A.3
BACKWARD 2	Multiplicative comparison word problems	4.OA.A.2
BACKWARD 3	Number patterns that follow a rule	4.OA.C.5
FORWARD 1	Write and solve equations for word problems	7.EE.B.4.a
FORWARD 2	Model a linear relationship with a function	8.F.B.4

SET 50 • 6.EE.B.6

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost \$8 each.

Write an expression for the cost of t tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q1/5

1840

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

8t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q1/5

Kim is k years old. Sam is 3 years older than Kim.

Write an expression for Sam's age.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q2/5

1841

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

$k + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q2/5

A pizza is cut into s slices. 4 friends share the slices equally.

Which expression shows how many slices each friend gets?

A. $4s$

B. $s - 4$

C. $\frac{s}{4}$

D. $s + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q3/5

1842

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $s/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q3/5

The length of a rectangle is 5 inches more than its width w .

Write an expression for the length.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q4/5

1843

QUESTION ID

S50-M-Q4

TYPE

Short answer

CORRECT ANSWER

$w + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q4/5

A gym charges \$20 to join plus \$15 per month.
Write an expression for the total cost after m months.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q5/5

1844

QUESTION ID

S50-M-Q5

TYPE

Short answer

CORRECT ANSWER

20 + 15m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q5/5

SET 50 • 6.EE.B.6

BACKWARD 1

Word problems with an unknown number • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 bags with the same number of apples.
There are 42 apples in all.
How many apples are in each bag?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q1/5

1846

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q1/5

There are 5 boxes with $\square$ pencils in each box. There are 35 pencils in all.

Which equation matches?

A. $35 \times 5 = \square$

B. $\square - 5 = 35$

C. $5 + \square = 35$

D. $5 \times \square = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q2/5

1847

QUESTION ID

S50-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $5 \times \square = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q2/5

True or false?

If $4 \times \square = 28$, then $\square = 7$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q3/5

1848

QUESTION ID

S50-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q3/5

There are 8 rows with $\square$ chairs in each row. There are 56 chairs in all.

What number is $\square$?

$\square =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q4/5

1849

QUESTION ID

S50-B1-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q4/5

Jan split 45 stickers equally among 9 friends.

How many stickers did each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q5/5

1850

QUESTION ID

S50-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 6.EE.B.6

BACKWARD 2

Multiplicative comparison word problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A red ribbon is 7 inches long. A blue ribbon is 3 times as long.

How long is the blue ribbon?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q1/5

1852

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

21 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q1/5

Ben has 6 books. Ali has 4 times as many books as Ben.

Which equation finds b , the number of books Ali has?

A. $b = 4 \times 6$

B. $b = 4 + 6$

C. $b = 6 \div 4$

D. $b = 6 - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q2/5

1853

QUESTION ID

S50-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $b = 4 \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q2/5

True or false?

A cat weighs 5 pounds. A dog weighs 8 times as much. The dog weighs 40 pounds.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q3/5

1854

QUESTION ID

S50-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q3/5

A bush is 4 feet tall. A tree is 5 times as tall.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q4/5

1855

QUESTION ID

S50-B2-Q4

TYPE

Short answer

CORRECT ANSWER

20 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q4/5

Sara ran 24 miles. That is 3 times as far as Tom ran.

How far did Tom run?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q5/5

1856

QUESTION ID

S50-B2-Q5

TYPE

Short answer

CORRECT ANSWER

8 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 6.EE.B.6

BACKWARD 3

Number patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking
• 4.OA.C.5

The rule is "add 4." The pattern starts at 3.

Write the first five numbers.

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q1/5

1858

QUESTION ID

S50-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3, 7, 11, 15, 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q1/5

The rule is "multiply by 2." The pattern starts at 1.

What is the 5th number?

- A. 8
- B. 10
- C. 16
- D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q2/5

1859

QUESTION ID

S50-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q2/5

True or false?

The rule "add 5," starting at 5, gives 5, 10, 15, 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q3/5

1860

QUESTION ID

S50-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q3/5

What is the rule for this pattern?

7, 11, 15, 19, ...

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q4/5

1861

QUESTION ID

S50-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Add 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q4/5

The rule is "add 6." The pattern starts at 2.

What is the 4th number?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q5/5

1862

QUESTION ID

S50-B3-Q5

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • 6.EE.B.6

FORWARD 1

Write and solve equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Sam has \$60. He spends \$8 each week.

Write and solve an equation to find when he will have \$20 left.

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q1/5

1864

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$60 - 8w = 20; w = 5$
weeks

GRADING NOTE Any equivalent equation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q1/5

A rectangle has a perimeter of 50 cm and a length of 15 cm.

Which equation can be used to find its width w ?

A. $2(15 + w) = 50$

B. $2(15w) = 50$

C. $15 + 2w = 50$

D. $15 + w = 50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q2/5

1865

QUESTION ID

S50-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $2(15 + w) = 50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q2/5

A taxi charges \$3 plus \$2 per mile. The fare was \$17.

How many miles was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q3/5

1866

QUESTION ID

S50-F1-Q3

TYPE

Short answer

CORRECT ANSWER

7 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q3/5

You buy 4 notebooks and a \$6 backpack. You spend \$18 in all.

How much does each notebook cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q4/5

1867

QUESTION ID

S50-F1-Q4

TYPE

Short answer

CORRECT ANSWER

\$3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q4/5

True or false?

"Five less than twice a number is 17" can be written as $2n - 5 = 17$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q5/5

1868

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q5/5

SET 50 • 6.EE.B.6

FORWARD 2

Model a linear relationship with a function • Grade 8 • Functions • 8.F.B.4

A plant is 4 cm tall. It grows 1.5 cm each week.

Write a function for its height h after w weeks.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q1/5

1870

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$h = 4 + 1.5w$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q1/5

Which situation matches $y = 2x + 10$, where y is dollars in savings after x weeks?

- A. Save \$12 each week.
- B. Start with \$10 and spend \$2 each week.
- C. Start with \$10 and save \$2 each week.
- D. Start with \$2 and save \$10 each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q2/5

1871

QUESTION ID

S50-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Start with \$10 and save \$2 each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q2/5

The table shows savings over time.

Write a function for the savings y after x weeks.

Weeks (x)	0	1	2
Savings (y)	30	45	60

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q3/5

1872

QUESTION ID

S50-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$$y = 15x + 30$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q3/5

A rental truck costs \$50 plus \$0.25 per mile.

Write a function for the cost C of driving m miles.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q4/5

1873

QUESTION ID

S50-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$C = 50 + 0.25m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

A line through (0, 5) and (2, 9) has an initial value of 5 and a rate of change of 2.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q5/5

1874

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q5/5

Solve one-step equations of the form

$$x + p = q$$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Subtraction as an unknown-addend problem	1.OA.B.4
BACKWARD 2	Adding and subtracting decimals	5.NBT.B.7
BACKWARD 3	Adding and subtracting fractions and mixed numbers	5.NF.A.1
FORWARD 1	Solve two-step equations	7.EE.B.4.a
FORWARD 2	Solve equations with variables on both sides	8.EE.C.7.b

SET 51 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 7.5 = 12$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q1/5

1877

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q1/5

Solve for n.

$$n - \frac{3}{4} = 5\frac{1}{2}$$

A. $4\frac{3}{4}$

B. $5\frac{1}{4}$

C. $6\frac{1}{4}$

D. $6\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q2/5

1878

QUESTION ID

S51-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 6 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q2/5

Solve for y.

$$y + 2\frac{1}{3} = 6$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q3/5

1879

QUESTION ID

S51-M-Q3

TYPE

Short answer

CORRECT ANSWER

3 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q3/5

Solve for m.

$$18.25 + m = 30$$

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q4/5

1880

QUESTION ID

S51-M-Q4

TYPE

Short answer

CORRECT ANSWER

11.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q4/5

Solve for k.

$$k - 4.8 = 9.2$$

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q5/5

1881

QUESTION ID

S51-M-Q5

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q5/5

SET 51 • 6.EE.B.7

BACKWARD 1

Subtraction as an unknown-addend problem • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4

True or false?

$12 - 7 = \square$ can be solved by thinking

$7 + \square = 12$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q1/5

1883

QUESTION ID

S51-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q1/5

What number goes in the box?

$$7 + \square = 15$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q2/5

1884

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q2/5

To find $16 - 9$, think $9 + \square = 16$.

What number goes in the box?

- A. 6
- B. 7
- C. 8
- D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q3/5

1885

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q3/5

What number goes in the box?

$$\square + 8 = 13$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q4/5

1886

QUESTION ID

S51-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q4/5

True or false?

To find $11 - 5$, you can think $5 + 6 = 11$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q5/5

1887

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 6.EE.B.7

BACKWARD 2

Adding and subtracting decimals • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$12 - 7.5$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q1/5

1889

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q1/5

1889

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$30 - 18.25$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q2/5

1890

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

11.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q2/5

1890

True or false?

$$9.2 + 4.8 = 14$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q3/5

1891

QUESTION ID

S51-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q3/5

$$6.25 + 2.8 = ?$$

- A.** 6.53
- B.** 8.33
- C.** 9.05
- D.** 9.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q4/5

1892

QUESTION ID

S51-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$3.7 + 5.45$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q5/5

1893

QUESTION ID

S51-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q5/5

1893

SET 51 • 6.EE.B.7

BACKWARD 3

Adding and subtracting fractions and mixed numbers • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$6 - 2\frac{1}{3}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q1/5

1895

QUESTION ID

S51-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q1/5

Add.

$$5\frac{1}{2} + \frac{3}{4}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q2/5

1896

QUESTION ID

S51-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q2/5

True or false?

$$\frac{2}{3} + \frac{1}{6} = \frac{5}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q3/5

1897

QUESTION ID

S51-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q3/5

$$\frac{7}{8} - \frac{1}{4} = ?$$

A. $\frac{1}{2}$

B. $\frac{5}{8}$

C. $\frac{3}{4}$

D. $\frac{6}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q4/5

1898

QUESTION ID

S51-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q4/5

Add.

$$1\frac{2}{5} + \frac{1}{2}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q5/5

1899

QUESTION ID

S51-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1 9/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • 6.EE.B.7

FORWARD 1

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 8 = 29$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q1/5

1901

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Solve for x.

$$-2x + 5 = 17$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q2/5

1902

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q2/5

1902

Solve for x.

$$\frac{x}{4} - 3 = 2$$

- A. -4
- B. $\frac{5}{4}$
- C. 5
- D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q3/5

1903

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q3/5

Solve for x.

$$0.6x + 1.5 = 4.5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q4/5

1904

QUESTION ID

S51-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q4/5

Solve for x.

$$5(x + 2) = 45$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q5/5

1905

QUESTION ID

S51-F1-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q5/5

SET 51 • 6.EE.B.7

FORWARD 2

Solve equations with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$7x - 4 = 3x + 12$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q1/5

1907

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q1/5

Solve for x.

$$-5x + 9 = x - 3$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q2/5

1908

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q2/5

Solve for x.

$$2.5x + 4 = 1.5x + 9$$

- A.** -5
- B.** 5
- C.** 6.5
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q3/5

1909

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{1}{2}x + 6 = 2x - 3$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q4/5

1910

QUESTION ID

S51-F2-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q4/5

Solve for x.

$$3(2x - 5) = 4x + 1$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q5/5

1911

QUESTION ID

S51-F2-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q5/5

Solve one-step equations of the form

$$px = q$$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Dividing by a fraction	6.NS.A.1
BACKWARD 3	Dividing decimals	5.NBT.B.7
FORWARD 1	Solve equations of the form $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solve multi-step linear equations	8.EE.C.7.b

SET 52 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$4x = 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q1/5

1914

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q1/5

Solve for y.

$$\frac{2}{3}y = 12$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q2/5

1915

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q2/5

Solve for m.

$$0.6m = 4.2$$

$$m = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q3/5

1916

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q3/5

Solve for n.

$$5n = 3.5$$

- A. -1.5
- B. 0.7
- C. 1.43
- D. 17.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q4/5

1917

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q4/5

Solve for k.

$$\frac{3}{4}k = \frac{9}{10}$$

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q5/5

1918

QUESTION ID

S52-M-Q5

TYPE

Short answer

CORRECT ANSWER

6/5

GRADING NOTE 1 1/5 or 1.2 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q5/5

SET 52 • 6.EE.B.7

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $48 \div 6$ by thinking $6 \times \square = 48$.

$\square =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q1/5

1920

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q1/5

True or false?

$63 \div 7 = 9$ because $7 \times 9 = 63$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q2/5

1921

QUESTION ID

S52-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q2/5

Which fact helps you find $45 \div 5$?

- A. $9 - 5 = 4$
- B. $45 \times 5 = 225$
- C. $5 \times 9 = 45$
- D. $5 + 40 = 45$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q3/5

1922

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $5 \times 9 = 45$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q3/5

What number goes in the box?

$$\square \times 6 = 42$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q4/5

1923

QUESTION ID

S52-B1-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q4/5

True or false?

$24 \div 4 = 5$ because $4 \times 5 = 24$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q5/5

1924

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 6.EE.B.7

BACKWARD 2

Dividing by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$12 \div \frac{2}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q1/5

1926

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q1/5

Divide.

$$\frac{9}{10} \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q2/5

1927

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q2/5

True or false?

$$5 \div \frac{5}{8} = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q3/5

1928

QUESTION ID

S52-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q3/5

$$\frac{9}{10} \div \frac{3}{5} = ?$$

A. $\frac{27}{50}$

B. $\frac{2}{3}$

C. $1\frac{1}{3}$

D. $1\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q4/5

1929

QUESTION ID

S52-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q4/5

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q5/5

1930

QUESTION ID

S52-B2-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 6.EE.B.7

BACKWARD 3

Dividing decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$4.2 \div 0.6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q1/5

1932

QUESTION ID

S52-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$30 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q2/5

1933

QUESTION ID

S52-B3-Q2

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q2/5

1933

True or false?

$$3.5 \div 5 = 0.7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q3/5

1934

QUESTION ID

S52-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q3/5

$$2.4 \div 0.3 = ?$$

- A.** 0.8
- B.** 7
- C.** 8
- D.** 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q4/5

1935

QUESTION ID

S52-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7.2 \div 9$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q5/5

1936

QUESTION ID

S52-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q5/5

1936

SET 52 • 6.EE.B.7

FORWARD 1

Solve equations of the form $p(x + q) = r$ • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Solve for x.

$$3(x - 4) = 21$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q1/5

1938

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q1/5

Solve for x.

$$\frac{1}{2}(x + 6) = 9$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q2/5

1939

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q2/5

Solve for x.

$$3(x + 1.5) = 12$$

- A.** 2
- B.** 2.5
- C.** 4
- D.** 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q3/5

1940

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q3/5

Solve for x.

$$-2(x - 3) = 14$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q4/5

1941

QUESTION ID

S52-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q4/5

Solve for x.

$$0.5(x + 8) = 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q5/5

1942

QUESTION ID

S52-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q5/5

SET 52 • 6.EE.B.7

FORWARD 2

Solve multi-step linear equations • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$3(2x - 1) = 4x + 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q1/5

1944

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{2}{3}(x + 3) = x - 1$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q2/5

1945

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q2/5

Solve for x.

$$5x - 2(x + 1) = 10$$

A. 2

B. $\frac{8}{3}$

C. 4

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q3/5

1946

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q3/5

Solve for x.

$$0.4(10x - 5) = 2x + 4$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q4/5

1947

QUESTION ID

S52-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q4/5

Solve for x.

$$6x = 2(x + 8)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q5/5

1948

QUESTION ID

S52-F2-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q5/5

Write and solve one-step equations for real-world problems

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Word problems with an unknown factor	3.OA.A.3
BACKWARD 2	Word problems with an unknown addend	2.OA.A.1
BACKWARD 3	A letter stands for an unknown quantity	3.OA.D.8
FORWARD 1	Write and solve two-step equations for word problems	7.EE.B.4.a
FORWARD 2	Solve real-world problems with variables on both sides	8.EE.C.7.b

SET 53 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

A pack of 6 pens costs \$4.50.

Write and solve an equation to find the cost p of one pen.

Equation: _____

$p =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q1/5

1951

QUESTION ID

S53-M-Q1

TYPE

Short answer

CORRECT ANSWER

$6p = 4.50; p = \$0.75$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q1/5

After spending \$12.75, Sam has \$30.25 left.

Write and solve an equation to find how much money m Sam started with.

Equation: _____

$m =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q2/5

1952

QUESTION ID

S53-M-Q2

TYPE

Short answer

CORRECT ANSWER

$$m - 12.75 = 30.25;$$

$$m = \$43$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q2/5

A board is cut into 5 equal pieces. Each piece is 1.8 meters long.

Write and solve an equation to find the length L of the board.

Equation: _____

$L =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q3/5

1953

QUESTION ID

S53-M-Q3

TYPE

Short answer

CORRECT ANSWER

**$L \div 5 = 1.8$; $L = 9$
meters**

GRADING NOTE $L = 5 \times 1.8$ is also acceptable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q3/5

Maya is 1.35 m tall. She is 0.4 m taller than her brother.
Which equation can be used to find her brother's height b ?

A. $0.4b = 1.35$

B. $1.35 + 0.4 = b$

C. $b - 0.4 = 1.35$

D. $b + 0.4 = 1.35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q4/5

1954

QUESTION ID

S53-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $b + 0.4 = 1.35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q4/5

A recipe uses $\frac{3}{4}$ cup of sugar per batch. Kim used 6 cups of sugar.

Write and solve an equation to find the number of batches b .

Equation: _____

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q5/5

1955

QUESTION ID

S53-M-Q5

TYPE

Short answer

CORRECT ANSWER

**$(\frac{3}{4})b = 6; b = 8$
batches**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q5/5

SET 53 • 6.EE.B.7

BACKWARD 1

Word problems with an unknown factor • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

4 friends share some grapes equally. Each gets 9 grapes.

How many grapes were there?

Grapes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q1/5

1957

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q1/5

There are 7 teams with $\square$ players each. There are 56 players in all.

Which equation matches?

A. $7 \times \square = 56$

B. $56 \times 7 = \square$

C. $\square - 7 = 56$

D. $7 + \square = 56$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q2/5

1958

QUESTION ID

S53-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $7 \times \square = 56$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q2/5

True or false?

If $3 \times \square = 24$, then $\square = 8$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q3/5

1959

QUESTION ID

S53-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q3/5

A teacher puts 40 books into 5 equal stacks.

How many books are in each stack?

Books: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q4/5

1960

QUESTION ID

S53-B1-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q4/5

Each box holds 6 eggs. There are 54 eggs.

How many boxes are there?

Boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q5/5

1961

QUESTION ID

S53-B1-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 6.EE.B.7

BACKWARD 2

Word problems with an unknown addend • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Lin had some stickers. She got 15 more. Now she has 42.

How many did she have at first?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q1/5

1963

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q1/5

A plant grew 8 cm and is now 35 cm tall.

Which equation finds the starting height h ?

- A. $8h = 35$
- B. $h - 8 = 35$
- C. $h + 35 = 8$
- D. $h + 8 = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q2/5

1964

QUESTION ID

S53-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $h + 8 = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q2/5

True or false?

Tom spent \$17 and has \$26 left, so he started with \$43.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q3/5

1965

QUESTION ID

S53-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q3/5

A bus had 30 people. Some got off. Now there are 18.

How many got off?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q4/5

1966

QUESTION ID

S53-B2-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q4/5

Ada read 24 pages on Monday and some more on Tuesday. She read 61 pages in all.

How many pages did she read on Tuesday?

Pages: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q5/5

1967

QUESTION ID

S53-B2-Q5

TYPE

Short answer

CORRECT ANSWER

37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 6.EE.B.7

BACKWARD 3

A letter stands for an unknown quantity • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

True or false?

In $5 \times n = 40$, the letter n stands for 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q1/5

1969

QUESTION ID

S53-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q1/5

In the equation $p + 9 = 20$, what number is p ?

$p =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q2/5

1970

QUESTION ID

S53-B3-Q2

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q2/5

Which equation matches "a number plus 6 is 15"?

A. $n + 15 = 6$

B. $n - 6 = 15$

C. $6n = 15$

D. $n + 6 = 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q3/5

1971

QUESTION ID

S53-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $n + 6 = 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q3/5

In the equation $3 \times k = 27$, what number is k ?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q4/5

1972

QUESTION ID

S53-B3-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q4/5

True or false?

In $m - 4 = 10$, the letter m stands for 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q5/5

1973

QUESTION ID

S53-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • 6.EE.B.7

FORWARD 1

Write and solve two-step equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a \$25 fee plus \$15 per month. Jen paid \$130.

Write and solve an equation to find the number of months m .

$m =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q1/5

1975

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$25 + 15m = 130;$
 $m = 7$ months**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q1/5

A plumber charges \$45 plus \$60 per hour. A bill is \$225.

How many hours did the plumber work?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q2/5

1976

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q2/5

A class buys 12 tickets and pays a \$15 group fee. The total is \$159.

Which equation finds the cost t of one ticket?

- A.** $12t - 15 = 159$ **B.** $12t + 15 = 159$
C. $12(t + 15) = 159$ **D.** $15t + 12 = 159$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q3/5

1977

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $12t + 15 = 159$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q3/5

Mo saves \$18 per week and already has \$54. He wants \$180.

How many weeks w will it take?

$W =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q4/5

1978

QUESTION ID

S53-F1-Q4

TYPE

Short answer

CORRECT ANSWER

7 weeks

GRADING NOTE $54 + 18w = 180$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q4/5

True or false?

A \$6 pizza plus 3 toppings costs \$10.50. Each topping costs \$1.50.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q5/5

1979

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q5/5

SET 53 • 6.EE.B.7

FORWARD 2

Solve real-world problems with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges \$20 plus \$3 per shirt. Store B charges \$5 per shirt.

Write and solve an equation to find when the costs are equal.

Shirts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q1/5

1981

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**$20 + 3s = 5s$; $s = 10$
shirts**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q1/5

Ana has \$120 and spends \$8 per week. Ben has \$40 and saves \$12 per week.

After how many weeks will they have the same amount?

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q2/5

1982

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 weeks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q2/5

Taxi A charges \$4 plus \$2.50 per mile. Taxi B charges \$1 plus \$3 per mile.

For how many miles do the rides cost the same?

A. 3

B. 5

C. 6

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q3/5

1983

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q3/5

A candle 30 cm tall burns 1.5 cm per hour. Another candle 20 cm tall burns 0.5 cm per hour.

After how many hours are they the same height?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q4/5

1984

QUESTION ID

S53-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q4/5

Plan A: \$15 per month plus \$0.10 per text. Plan B: \$0.25 per text.

For how many texts do the plans cost the same?

Texts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q5/5

1985

QUESTION ID

S53-F2-Q5

TYPE

Short answer

CORRECT ANSWER

100 texts

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q5/5

Write an inequality $x > c$ or $x < c$ for a real-world condition

MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Inequality symbols	1.NBT.B.3
BACKWARD 2	Meaning of "at least," "at most," "more than," and "fewer than"	6.NS.C.7.b
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Write and solve inequalities for real-world problems	7.EE.B.4.b
FORWARD 2	Use two linear equations to decide when one quantity is greater...	8.EE.C.8.c

SET 54 • 6.EE.B.8

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Riders must be at least 48 inches tall.

Write an inequality for the height h of a rider.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q1/5

1988

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

$h \geq 48$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q1/5

A bag can hold at most 20 pounds.

Which inequality describes the weight w the bag can hold?

A. $w \geq 20$

B. $w < 20$

C. $w \leq 20$

D. $w > 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q2/5

1989

QUESTION ID

S54-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $w \leq 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q2/5

The temperature must stay below -3°C .

Write an inequality for the temperature t .

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q3/5

1990

QUESTION ID

S54-M-Q3

TYPE

Short answer

CORRECT ANSWER

$t < -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q3/5

More than 25 people must sign up for a trip to happen.

Write an inequality for the number of people p .

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q4/5

1991

QUESTION ID

S54-M-Q4

TYPE

Short answer

CORRECT ANSWER

$p > 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q4/5

Children 12 years old and younger get a discount.

Write an inequality for the age a of a child who gets the discount.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q5/5

1992

QUESTION ID

S54-M-Q5

TYPE

Short answer

CORRECT ANSWER

$a \leq 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q5/5

SET 54 • 6.EE.B.8

BACKWARD 1

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

58 ____ 85

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q1/5

1994

QUESTION ID

S54-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. <

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q1/5

True or false?

$30 > 29$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q2/5

1995

QUESTION ID

S54-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q2/5

Write $>$, $<$, or $=$ to compare.

64 ____ 46

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q3/5

1996

QUESTION ID

S54-B1-Q3

TYPE

Short answer

CORRECT ANSWER

$>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q3/5

Which symbol means "is greater than"?

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q4/5

1997

QUESTION ID

S54-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. >

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q4/5

True or false?

$$17 < 12$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q5/5

1998

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 6.EE.B.8

BACKWARD 2

Meaning of "at least," "at most," "more than," and "fewer than" • Grade 6 • The Number System • 6.NS.C.7.b

True or false?

"At least 10" includes the number 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q1/5

2000

QUESTION ID

S54-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q1/5

What does "at most 6" mean?

- A.** 6 or more
- B.** Less than 6
- C.** More than 6
- D.** 6 or less

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q2/5

2001

QUESTION ID

S54-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 6 or less

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q2/5

True or false?

"Fewer than 8" includes the number 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q3/5

2002

QUESTION ID

S54-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q3/5

Which number is NOT "at least 12"?

- A.** 11
- B.** 12
- C.** 13
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q4/5

2003

QUESTION ID

S54-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q4/5

True or false?

9 is "more than 7."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q5/5

2004

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 6.EE.B.8

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 3 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q1/5

2006

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q1/5

True or false?

-10°C is colder than -3°C .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q2/5

2007

QUESTION ID

S54-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q2/5

Which number represents 15 feet below sea level?

- A.** -15
- B.** -1.5
- C.** 0
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q3/5

2008

QUESTION ID

S54-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q3/5

Write an integer for a temperature of 7 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q4/5

2009

QUESTION ID

S54-B3-Q4

TYPE

Short answer

CORRECT ANSWER

-7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q4/5

True or false?

"Below -3°C " includes -5°C .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q5/5

2010

QUESTION ID

S54-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • 6.EE.B.8

FORWARD 1

Write and solve inequalities for real-world problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

You have \$50. Each ride costs \$6.

Write and solve an inequality to find how many rides r you can afford.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q1/5

2012

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$6r \leq 50$; $r \leq 8 \frac{1}{3}$, so
at most 8 rides**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q1/5

A salesperson earns \$80 plus \$12 for each sale.

Write and solve an inequality for the number of sales s needed to earn more than \$200.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q2/5

2013

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

**$80 + 12s > 200;$
 $s > 10$ (at least 11 sales)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q2/5

A phone has 75% battery and uses 5% per hour. It must stay above 20%.

Which inequality gives the hours h ?

A. $75 - 5h > 20$

B. $75 + 5h > 20$

C. $75 - 5h < 20$

D. $5h - 75 > 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q3/5

2014

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $75 - 5h > 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q3/5

A van can carry at most 1,200 pounds. The driver weighs 180 pounds. Each box weighs 60 pounds. Write and solve an inequality for the number of boxes b .

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q4/5

2015

QUESTION ID

S54-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**$180 + 60b \leq 1,200;$
 $b \leq 17$ (at most 17
boxes)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q4/5

True or false?

The solution of $3x + 2 \geq 17$ is $x \geq 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q5/5

2016

QUESTION ID

S54-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q5/5

FORWARD 2

Use two linear equations to decide when one quantity is greater • Grade 8 •
Expressions & Equations • 8.EE.C.8.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Plan A costs \$20 plus \$5 per month. Plan B costs \$35 plus \$2 per month.

After how many months does Plan A cost more than Plan B?

Months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

2018

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**After 5 months
($m > 5$)**

GRADING NOTE $20 + 5m > 35 + 2m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

Tank A has 100 gallons and drains 6 gallons per minute.
Tank B has 40 gallons and fills 4 gallons per minute.
After how many minutes does Tank B hold more water?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

2019

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

**After 6 minutes
($t > 6$)**

GRADING NOTE $40 + 4t > 100 - 6t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

Sara has \$50 and saves \$8 per week. Tom has \$90 and saves \$3 per week.

When will Sara have more money than Tom?

- A.** After 8 weeks **B.** After 3 weeks
C. After 5 weeks **D.** Never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

2020

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. After 8 weeks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

Gym X charges \$60 plus \$4 per class. Gym Y charges \$12 per class.

For how many classes is Gym X cheaper?

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

2021

QUESTION ID

S54-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**8 or more classes
($c > 7.5$)**

GRADING NOTE $60 + 4c < 12c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

$y = 3x + 2$ and $y = x + 10$ are equal when $x = 4$.
For $x > 4$, $3x + 2$ is greater.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

2022

QUESTION ID

S54-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

Represent solutions of $x > c$ or $x < c$ on a number line

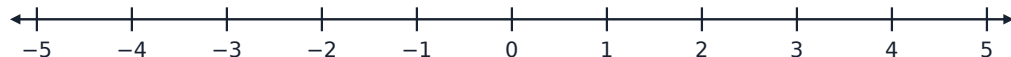
MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Locating numbers on a number line	6.NS.C.6.c
BACKWARD 2	Greater numbers are to the right	6.NS.C.7.a
BACKWARD 3	Meaning of inequality symbols	1.NBT.B.3
FORWARD 1	Solve and graph two-step inequalities	7.EE.B.4.b
FORWARD 2	Linear equations with one, no, or infinitely many solutions — near...	8.EE.C.7.a

SET 55 • 6.EE.B.8

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Graph the solutions of $x > 3$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q1/5

2025

QUESTION ID

S55-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

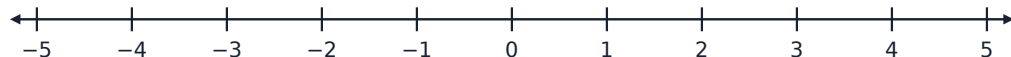
**Open circle at 3 with
the line shaded to
the right**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q1/5

Graph the solutions of $x \leq -1$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q2/5

2026

QUESTION ID

S55-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

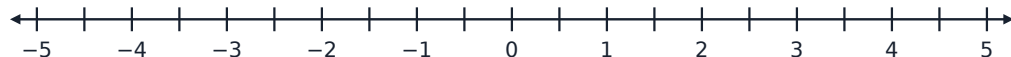
**Closed circle at -1
with the line shaded
to the left**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q2/5

Graph the solutions of $x < 2.5$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q3/5

2027

QUESTION ID

S55-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

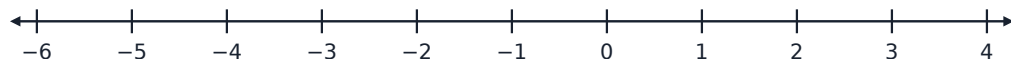
**Open circle at 2.5
with the line shaded
to the left**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q3/5

Graph the solutions of $x \geq -4$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q4/5

2028

QUESTION ID

S55-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

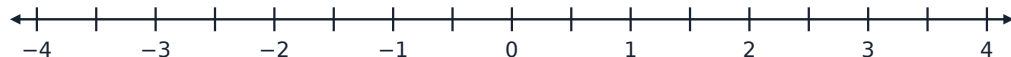
**Closed circle at -4
with the line shaded
to the right**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q4/5

Graph the solutions of $x > -1.5$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q5/5

2029

QUESTION ID

S55-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Open circle at -1.5
with the line shaded
to the right**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

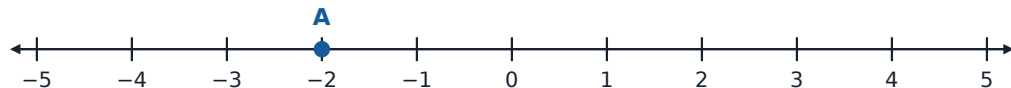
Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q5/5

SET 55 • 6.EE.B.8

BACKWARD 1

Locating numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q1/5

2031

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q1/5

Which point is at 3.5?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q2/5

2032

QUESTION ID

S55-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

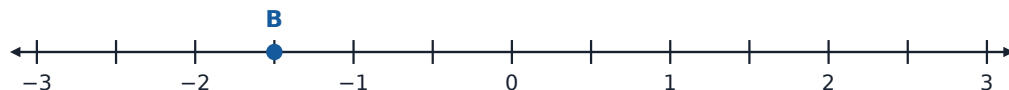
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q2/5

True or false?

Point B is at -1.5 .



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q3/5

2033

QUESTION ID

S55-B1-Q3

TYPE

True or false

CORRECT ANSWER

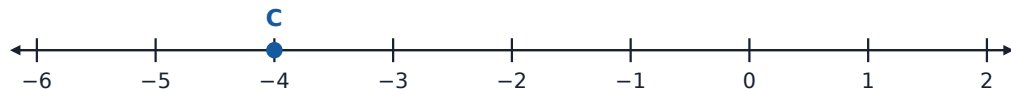
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q3/5

What number is at point C?



C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q4/5

2034

QUESTION ID

S55-B1-Q4

TYPE

Short answer

CORRECT ANSWER

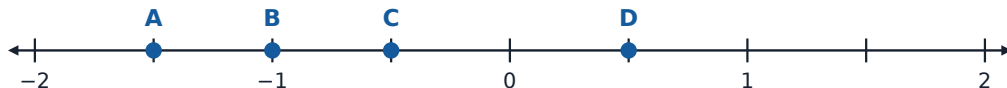
-4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q4/5

Which point is at $-\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q5/5

2035

QUESTION ID

S55-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 6.EE.B.8

BACKWARD 2

Greater numbers are to the right • Grade 6 • The Number System • 6.NS.C.7.a

True or false?

Every number to the right of 3 on a number line is greater than 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q1/5

2037

QUESTION ID

S55-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q1/5

Which numbers are less than -1 ?

- A.** Numbers to the right of -1
- B.** Numbers to the left of -1
- C.** Only -2
- D.** Only 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q2/5

2038

QUESTION ID

S55-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Numbers to the left of -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q2/5

True or false?

-3 is greater than -1 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q3/5

2039

QUESTION ID

S55-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q3/5

Is 2.5 to the left or the right of 2 on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q4/5

2040

QUESTION ID

S55-B2-Q4

TYPE

Short answer

CORRECT ANSWER

To the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q4/5

Which number is greater than -4 ?

- A.** -6
- B.** -5
- C.** -4.5
- D.** -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q5/5

2041

QUESTION ID

S55-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 6.EE.B.8

BACKWARD 3

Meaning of inequality symbols • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

What does the symbol $>$ mean?

- A. Is equal to
- B. Is greater than
- C. Is less than

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q1/5

2043

QUESTION ID

S55-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Is greater than

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q1/5

True or false?

The symbol $<$ means "is less than."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q2/5

2044

QUESTION ID

S55-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q2/5

Which statement is true?

- A.** $8 < 5$
- B.** $8 = 5$
- C.** $8 > 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q3/5

2045

QUESTION ID

S55-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $8 > 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q3/5

Write the symbol for "is less than."

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q4/5

2046

QUESTION ID

S55-B3-Q4

TYPE

Short answer

CORRECT ANSWER

<

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q4/5

True or false?

$12 > 21$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q5/5

2047

QUESTION ID

S55-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

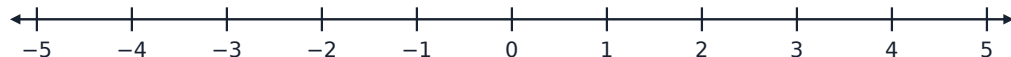
Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • 6.EE.B.8

FORWARD 1

Solve and graph two-step inequalities • Grade 7 • Expressions & Equations •
7.EE.B.4.b

Solve $2x + 1 > 7$. Then graph the solutions.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q1/5

2049

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

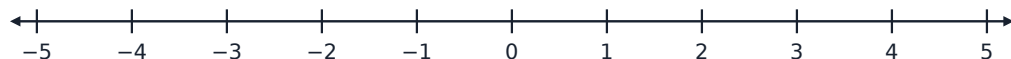
$x > 3$; open circle at 3, shaded to the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q1/5

Solve $4x - 3 \leq 5$. Then graph the solutions.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q2/5

2050

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

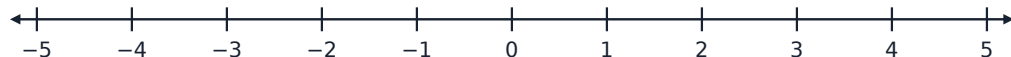
**$x \leq 2$; closed circle
at 2, shaded to the
left**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q2/5

Solve $-2x + 4 < 10$. Then graph the solutions.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q3/5

2051

QUESTION ID

S55-F1-Q3

TYPE

Short answer

CORRECT ANSWER

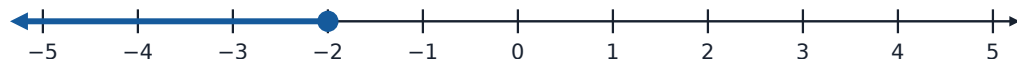
**$x > -3$; open circle
at -3 , shaded to the
right**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q3/5

Which inequality does the graph show?



A. $x < -2$

B. $x > -2$

C. $x \geq -2$

D. $x \leq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q4/5

2052

QUESTION ID

S55-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x \leq -2$

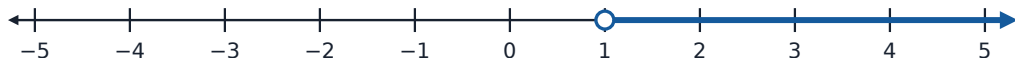
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q4/5

True or false?

The graph shows the solutions of $3x + 2 > 5$.



- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q5/5

2053

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q5/5

FORWARD 2

Linear equations with one, no, or infinitely many solutions • Grade 8 •
Expressions & Equations • 8.EE.C.7.a

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

True or false?

$x + 3 = x + 3$ has infinitely many solutions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q1/5 • NEAREST RELATED

2055

QUESTION ID

S55-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q1/5 • NEAREST RELATED

How many solutions does the equation have?

$$2x + 5 = 2x - 1$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q2/5 • NEAREST RELATED

2056

QUESTION ID

S55-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which equation has infinitely many solutions?

- A. $4x - 1 = x$
- B. $4x = 0$
- C. $4(x - 1) = 4x - 4$
- D. $4(x - 1) = 4x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q3/5 • NEAREST RELATED

2057

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $4(x - 1) = 4x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q3/5 • NEAREST RELATED

How many solutions does the equation have?

$$3x - 7 = 3x - 7$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q4/5 • NEAREST RELATED

2058

QUESTION ID

S55-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q4/5 • NEAREST RELATED

How many solutions does the equation have?

$$6x - 2 = 4x + 8$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q5/5 • NEAREST RELATED

2059

QUESTION ID

S55-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q5/5 • NEAREST RELATED

Identify the independent and dependent variables

MAIN Grade 6 main question

6.EE.C.9

BACKWARD 1 Input-output rules

4.OA.C.5

BACKWARD 2 Two related number patterns

5.OA.B.3

FORWARD 1 Write an equation for a proportional relationship

7.RP.A.2.c

FORWARD 2 Functions: each input has exactly one output

8.F.A.1

SET 56 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

The equation $c = 2.5p$ gives the cost c of p pounds of grapes.

Which is the independent variable?

- A.** p (pounds)
- B.** c (cost)
- C.** 2.5
- D.** There is none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q1/5

2062

QUESTION ID

S56-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. p (pounds)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q1/5

Money earned m depends on the number of hours worked h .

Which is the dependent variable?

Dependent variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q2/5

2063

QUESTION ID

S56-M-Q2

TYPE

Short answer

CORRECT ANSWER

m (money earned)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q2/5

A plant's height depends on the number of days it grows.

Which is the independent variable?

- A.** The plant's height **B.** The number of days
C. Neither **D.** Both

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q3/5

2064

QUESTION ID

S56-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The number of days

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q3/5

In $y = 4x + 3$, which variable is the dependent variable?

Dependent variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q4/5

2065

QUESTION ID

S56-M-Q4

TYPE

Short answer

CORRECT ANSWER

y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q4/5

True or false?

Ice cream sales s depend on the temperature t .
So t is the independent variable.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q5/5

2066

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q5/5

SET 56 • 6.EE.C.9

BACKWARD 1

Input-output rules • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "multiply by 3."

What is the output when the input is 7?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q1/5

2068

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q1/5

True or false?

The rule is "add 6." When the input is 4, the output is 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q2/5

2069

QUESTION ID

S56-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q2/5

What is the rule?

In	2	4	6
Out	6	8	10

- A.** Multiply by 2 **B.** Multiply by 3
C. Add 2 **D.** Add 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q3/5

2070

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Add 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q3/5

The rule is "subtract 2."

What is the output when the input is 15?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q4/5

2071

QUESTION ID

S56-B1-Q4

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q4/5

True or false?

In an input-output table, the output depends on the input.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q5/5

2072

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 6.EE.C.9

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

True or false?

Pattern A: 0, 2, 4, 6

Pattern B: 0, 6, 12, 18

Each number in Pattern B is 3 times the matching number in Pattern A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q1/5

2074

QUESTION ID

S56-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q1/5

The rule is "add 5," starting at 0.

What number comes next? 0, 5, 10, ____

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q2/5

2075

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

When Pattern A is 12, what is the matching number in Pattern B?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q3/5

2076

QUESTION ID

S56-B2-Q3

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q3/5

Pattern A starts at 0 and adds 1. Pattern B starts at 0 and adds 4.

Which ordered pairs come from the patterns?

- A.** (0, 4), (1, 5), (2, 6) **B.** (0, 0), (1, 4), (2, 8)
C. (1, 4), (2, 5), (3, 6) **D.** (0, 0), (4, 1), (8, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q4/5

2077

QUESTION ID

S56-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (0, 0), (1, 4), (2, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q4/5

True or false?

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Each number in Pattern B is 5 more than the matching number in Pattern A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q5/5

2078

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 6.EE.C.9

FORWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Tomatoes cost \$3.20 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q1/5

2080

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$c = 3.2p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for the cost c of n ounces.

Ounces (n)	4	10
Cost (c)	\$1.20	\$3.00

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q2/5

2081

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$c = 0.3n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q2/5

Which equation represents a proportional relationship?

- A. $y = 4.5x$
- B. $y = 4.5x + 1$
- C. $y = x + 4.5$
- D. $y = 4.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q3/5

2082

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = 4.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q3/5

A car uses 3 gallons of gas to travel 84 miles.

Write an equation for the miles m traveled on g gallons.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q4/5

2083

QUESTION ID

S56-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$m = 28g$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q4/5

True or false?

$y = \frac{2}{3}x$ represents a proportional relationship.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q5/5

2084

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q5/5

SET 56 • 6.EE.C.9

FORWARD 2

Functions: each input has exactly one output • Grade 8 • Functions • 8.F.A.1

True or false?

A function assigns exactly one output to each input.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q1/5

2086

QUESTION ID

S56-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q1/5

Which table does NOT represent a function?

- A. x: 0, 1, 2 y: 0, 2, 4
- B. x: 1, 1, 2 y: 3, 5, 7
- C. x: 2, 3, 4 y: 5, 6, 7
- D. x: 1, 2, 3 y: 4, 4, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q2/5

2087

QUESTION ID

S56-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. x: 1, 1, 2 y: 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q2/5

In the function $C = 12t$, which variable is the input?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q3/5

2088

QUESTION ID

S56-F2-Q3

TYPE

Short answer

CORRECT ANSWER

t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q3/5

True or false?

The set of ordered pairs (1, 2), (2, 3), (1, 4) represents a function.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q4/5

2089

QUESTION ID

S56-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q4/5

A function gives the height of a ball after t seconds.
What is the input?

- A. The speed
- B. The ball
- C. The height
- D. The time t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q5/5

2090

QUESTION ID

S56-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The time t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q5/5

Write an equation for the relationship shown in a table or graph

MAIN	Grade 6 main question	6.EE.C.9
BACKWARD 1	Finding a rule from a table	4.OA.C.5
BACKWARD 2	Reading points on a graph	5.G.A.2
BACKWARD 3	Ordered pairs from two patterns	5.OA.B.3
FORWARD 1	Equations for proportional relationships from tables and graphs	7.RP.A.2.c
FORWARD 2	Write a linear function from a table or graph	8.F.B.4

SET 57 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation that shows how y depends on x .

x	1	2	3	4
y	6	12	18	24

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q1/5

2093

QUESTION ID

S57-M-Q1

TYPE

Short answer

CORRECT ANSWER

$y = 6x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

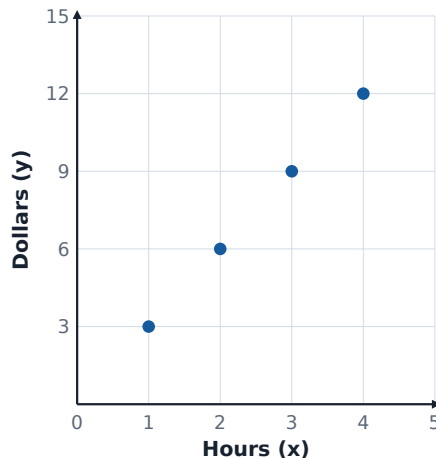
ID S57-M-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q1/5

The graph shows y dollars earned for x hours worked.

Write an equation for the relationship.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q2/5

2094

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q2/5

The table shows a plant's height h after w weeks.

Write an equation for h in terms of w .

Weeks (w)	0	1	2	3
Height (h)	5	7	9	11

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q3/5

2095

QUESTION ID

S57-M-Q3

TYPE

Short answer

CORRECT ANSWER

$$h = 2w + 5$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q3/5

Which equation matches the table?

x	1	2	3
y	4	5	6

A. $y = 4x$

B. $y = x + 4$

C. $y = x + 3$

D. $y = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q4/5

2096

QUESTION ID

S57-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $y = x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q4/5

The table shows the cost c of n bags of popcorn.

Write an equation for c in terms of n .

Bags (n)	2	4	6
Cost (c)	7	14	21

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q5/5

2097

QUESTION ID

S57-M-Q5

TYPE

Short answer

CORRECT ANSWER

$c = 3.5n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q5/5

SET 57 • 6.EE.C.9

BACKWARD 1

Finding a rule from a table • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

What is the rule?

In	1	2	3	4
Out	5	10	15	20

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q1/5

2099

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Multiply by 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q1/5

What is the rule?

In	3	5	7
Out	7	9	11

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q2/5

2100

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Add 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q2/5

True or false?

The rule for the table is "multiply by 4."

In	1	2	3
Out	4	8	12

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q3/5

2101

QUESTION ID

S57-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q3/5

What is the rule?

In	3	6	9
Out	1	2	3

- A.** Multiply by 3 **B.** Subtract 2
C. Add 3 **D.** Divide by 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q4/5

2102

QUESTION ID

S57-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Divide by 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q4/5

The rule is "multiply by 6." What is the output when the input is 5?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q5/5

2103

QUESTION ID

S57-B1-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q5/5

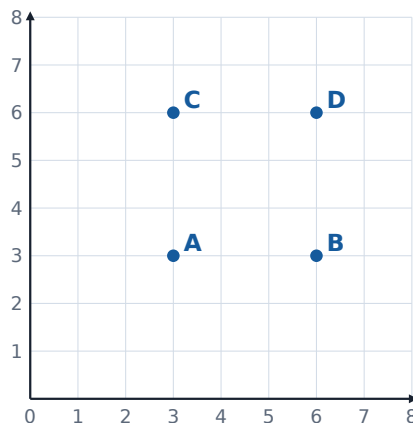
SET 57 • 6.EE.C.9

BACKWARD 2

Reading points on a graph • Grade 5 • Geometry • 5.G.A.2

Which point is at (3, 6)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q1/5

2105

QUESTION ID

S57-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

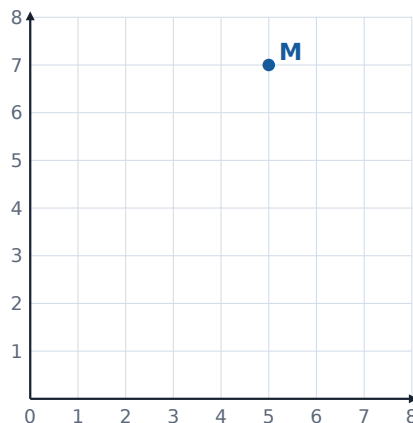
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q1/5

What are the coordinates of point M?

M = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q2/5

2106

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q2/5

True or false?

The point (2, 8) is 2 units right and 8 units up from the origin.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q3/5

2107

QUESTION ID

S57-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

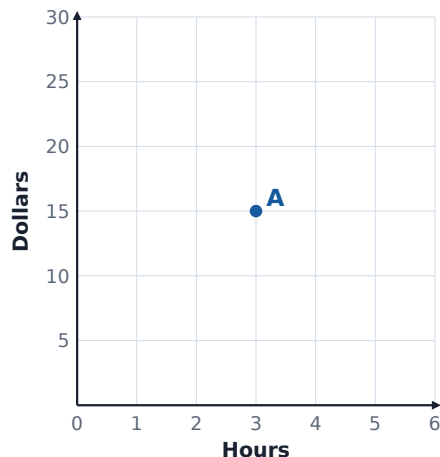
ID S57-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q3/5

The graph shows dollars earned after working some hours.

How many dollars were earned after 3 hours?

Dollars: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q4/5

2108

QUESTION ID

S57-B2-Q4

TYPE

Short answer

CORRECT ANSWER

\$15

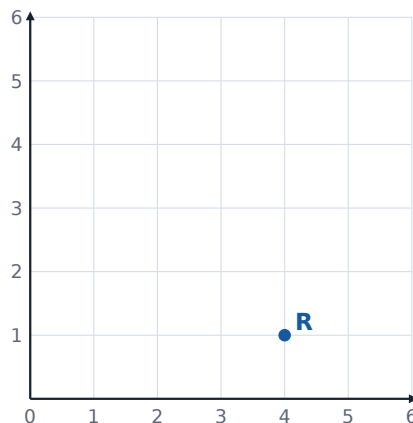
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q4/5

Which ordered pair names point R?

- A. (4, 4)
- B. (1, 4)
- C. (1, 1)
- D. (4, 1)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q5/5

2109

QUESTION ID

S57-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. (4, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 6.EE.C.9

BACKWARD 3

Ordered pairs from two patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Write the ordered pair formed by the 4th numbers.

Ordered pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q1/5

2111

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q1/5

True or false?

The pairs (1, 3), (2, 6), (3, 9) show that each y-value is 3 times the x-value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q2/5

2112

QUESTION ID

S57-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q2/5

Pattern A adds 2. Pattern B adds 6. Both start at 0.
How does each B number compare to its A number?

- A. 2 more
- B. 3 times as large
- C. 4 more
- D. 6 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q3/5

2113

QUESTION ID

S57-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q3/5

Pattern A: 0, 2, 4

Pattern B: 0, 14, 28

What is the rule that changes each A number into its B number?

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q4/5

2114

QUESTION ID

S57-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Multiply by 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q4/5

True or false?

The pairs (0, 0), (1, 4), (2, 8) come from Pattern A "add 4" and Pattern B "add 1."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q5/5

2115

QUESTION ID

S57-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • 6.EE.C.9

FORWARD 1

Equations for proportional relationships from tables and graphs • Grade 7 •
Ratios & Proportional Relationships • 7.RP.A.2.c

Write an equation for the proportional relationship in the table.

x	2	5	8
y	7	17.5	28

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q1/5

2117

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$y = 3.5x$

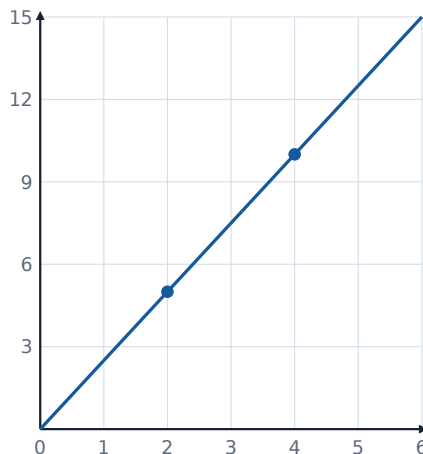
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q1/5

Write an equation for the proportional relationship in the graph.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q2/5

2118

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

 $y = 2.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q2/5

The table shows a proportional relationship.

Which equation matches?

x	4	8	12
y	3	6	9

A. $y = 0.75x$

B. $y = 1.33x$

C. $y = 3x$

D. $y = x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q3/5

2119

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = 0.75x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q3/5

Write an equation for the cost c of p pounds.

Pounds (p)	3	6	9
Cost (c)	7.50	15.00	22.50

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q4/5

2120

QUESTION ID

S57-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$c = 2.5p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q4/5

True or false?

The table shows the equation $y = 6x$.

x	1	2	4
y	6	12	24

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q5/5

2121

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q5/5

SET 57 • 6.EE.C.9

FORWARD 2

Write a linear function from a table or graph • Grade 8 • Functions • 8.F.B.4

Write a function for the table.

x	0	1	2	3
y	4	7	10	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q1/5

2123

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$$y = 3x + 4$$

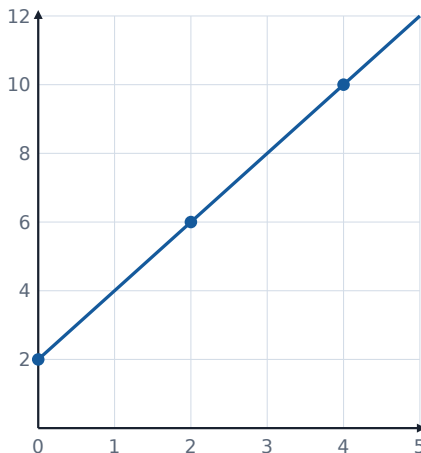
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q1/5

Write a function for the line in the graph.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q2/5

2124

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$y = 2x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q2/5

Which function matches the table?

x	0	1	2
y	10	8	6

A. $y = -2x + 10$

B. $y = 10x - 2$

C. $y = -2x$

D. $y = 2x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q3/5

2125

QUESTION ID

S57-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = -2x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q3/5

Write a function for the table.

x	1	2	3	4
y	9	13	17	21

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q4/5

2126

QUESTION ID

S57-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$y = 4x + 5$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q4/5

True or false?

The line through (0, 1) and (3, 7) has the equation $y = 2x + 1$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q5/5

2127

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q5/5

Area of right triangles and other triangles

MAIN

Grade 6 main question

6.G.A.1

BACKWARD 1

Area of rectangles

3.MD.C.7.b

BACKWARD 2

Half of a product

5.NF.B.4.a

BACKWARD 3

Right angles and perpendicular segments

4.G.A.1

FORWARD 1

Area and volume of figures made of triangles

7.G.B.6

FORWARD 2

Use the Pythagorean Theorem in right triangles

8.G.B.7

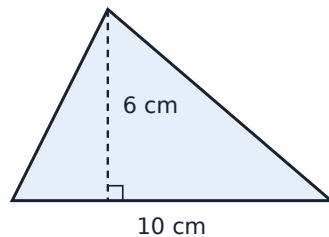
SET 58 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q1/5

2130

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

30 square cm

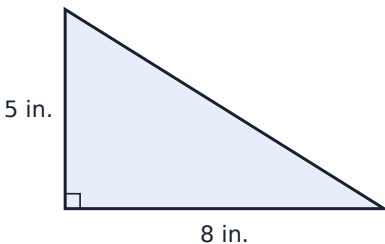
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q1/5

Find the area of the right triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q2/5

2131

QUESTION ID

S58-M-Q2

TYPE

Short answer

CORRECT ANSWER

20 square inches

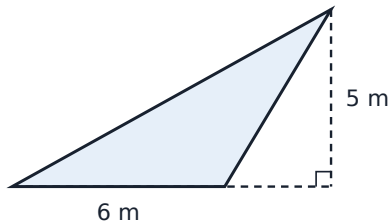
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q2/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q3/5

2132

QUESTION ID

S58-M-Q3

TYPE

Short answer

CORRECT ANSWER

15 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q3/5

A triangle has a base of 12 feet and a height of 7 feet.

What is its area?

- A. 19 square feet
- B. 38 square feet
- C. 42 square feet
- D. 84 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q4/5

2133

QUESTION ID

S58-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 42 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q4/5

2133

A triangular sign has a base of 3.5 feet and a height of 2 feet.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q5/5

2134

QUESTION ID

S58-M-Q5

TYPE

Short answer

CORRECT ANSWER

3.5 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q5/5

SET 58 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q1/5

2136

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

28 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q1/5

A rectangle is 9 feet long and 6 feet wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q2/5

2137

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

54 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q2/5

True or false?

A rectangle that is 5 cm long and 3 cm wide has an area of 15 square centimeters.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q3/5

2138

QUESTION ID

S58-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q3/5

A rectangle is 12 m long and 4 m wide.

What is its area?

- A. 16 square meters
- B. 24 square meters
- C. 32 square meters
- D. 48 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q4/5

2139

QUESTION ID

S58-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 48 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q4/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q5/5

2140

QUESTION ID

S58-B1-Q5

TYPE

Short answer

CORRECT ANSWER

60 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 6.G.A.1

BACKWARD 2

Half of a product • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times 10 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q1/5

2142

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q1/5

Multiply.

$$\frac{1}{2} \times 14$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q2/5

2143

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q2/5

True or false?

$$\frac{1}{2} \times 8 \times 5 = 20$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q3/5

2144

QUESTION ID

S58-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q3/5

$$\frac{1}{2} \times 12 \times 7 = ?$$

- A. 19
- B. 21
- C. 42
- D. 84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q4/5

2145

QUESTION ID

S58-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q4/5

Multiply.

$$\frac{1}{2} \times 9 \times 4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q5/5

2146

QUESTION ID

S58-B2-Q5

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 6.G.A.1

BACKWARD 3

Right angles and perpendicular segments • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q1/5

2148

QUESTION ID

S58-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q1/5

Which angle measure is a right angle?

- A. 45°
- B. 90°
- C. 100°
- D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q2/5

2149

QUESTION ID

S58-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

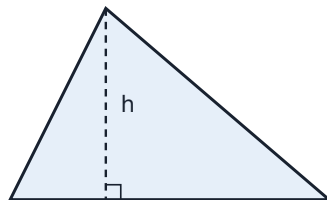
ID S58-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q2/5

True or false?

The dashed segment is perpendicular to the base of the triangle.

- A. True
- B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q3/5

2150

QUESTION ID

S58-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q3/5

How many degrees are in a right angle?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q4/5

2151

QUESTION ID

S58-B3-Q4

TYPE

Short answer

CORRECT ANSWER

90 degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q4/5

True or false?

A square has 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q5/5

2152

QUESTION ID

S58-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q5/5

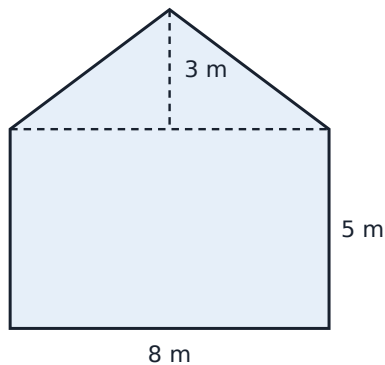
SET 58 • 6.G.A.1

FORWARD 1

Area and volume of figures made of triangles • Grade 7 • Geometry • 7.G.B.6

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q1/5

2154

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

52 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q1/5

A kite is made of two triangles. Each triangle has a base of 10 inches and a height of 4 inches.

What is the area of the kite?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q2/5

2155

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

40 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q2/5

A triangle with a base of 4 cm and a height of 3 cm is cut out of a rectangle that is 12 cm by 6 cm.

What is the area of the part that is left?

- A.** 60 square cm **B.** 66 square cm
C. 72 square cm **D.** 78 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q3/5

2156

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 66 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q3/5

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q4/5

2157

QUESTION ID

S58-F1-Q4

TYPE

Short answer

CORRECT ANSWER

120 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q4/5

True or false?

A figure made of a 6-by-4 rectangle and a triangle with a base of 3 and a height of 4 has an area of 30 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q5/5

2158

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q5/5

SET 58 • 6.G.A.1

FORWARD 2

Use the Pythagorean Theorem in right triangles • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.

What is the length of the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q1/5

2160

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 cm

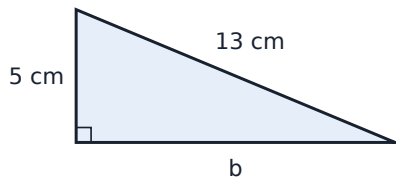
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q1/5

Find the length b .

$b =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q2/5

2161

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q2/5

A 10-foot ladder leans against a wall. The top reaches 8 feet up the wall.

How far is the bottom of the ladder from the wall?

- A.** 2 feet **B.** 6 feet
C. 12.8 feet **D.** 18 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q3/5

2162

QUESTION ID

S58-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q3/5

An isosceles triangle has a base of 16 inches and two sides of 10 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q4/5

2163

QUESTION ID

S58-F2-Q4

TYPE

Short answer

CORRECT ANSWER

48 square inches

GRADING NOTE Height = 6 in. by the Pythagorean Theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q4/5

A right triangle has legs of 9 m and 12 m.

What is the length of the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q5/5

2164

QUESTION ID

S58-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q5/5

Area of parallelograms and trapezoids

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area by splitting into rectangles	3.MD.C.7.d
BACKWARD 3	Classifying quadrilaterals	5.G.B.4
BACKWARD 4	Height is perpendicular to the base	4.G.A.1
FORWARD 1	Area of figures made of quadrilaterals in real-world problems	7.G.B.6
FORWARD 2	Use the Pythagorean Theorem to find a missing height	8.G.B.7

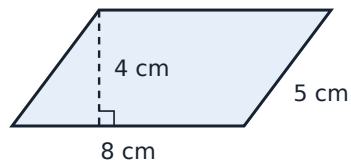
SET 59 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the parallelogram.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q1/5

2167

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

32 square cm

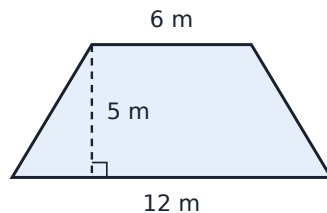
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q1/5

Find the area of the trapezoid.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q2/5

2168

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

45 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q2/5

A parallelogram has a base of 9 feet and a height of 5 feet.

What is its area?

- A.** 14 square feet **B.** 22.5 square feet
C. 28 square feet **D.** 45 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q3/5

2169

QUESTION ID

S59-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 45 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q3/5

A trapezoid has bases of 7 inches and 11 inches. Its height is 4 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q4/5

2170

QUESTION ID

S59-M-Q4

TYPE

Short answer

CORRECT ANSWER

36 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q4/5

A parallelogram has a base of 12.5 meters and a height of 4 meters.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q5/5

2171

QUESTION ID

S59-M-Q5

TYPE

Short answer

CORRECT ANSWER

50 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q5/5

SET 59 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q1/5

2173

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

24 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q1/5

A rectangle is 11 cm long and 5 cm wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q2/5

2174

QUESTION ID

S59-B1-Q2

TYPE

Short answer

CORRECT ANSWER

55 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q2/5

True or false?

A square with 6-meter sides has an area of 24 square meters.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q3/5

2175

QUESTION ID

S59-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q3/5

A rectangle is 15 inches long and 2 inches wide.

What is its area?

- A. 17 square inches
- B. 30 square inches
- C. 34 square inches
- D. 60 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q4/5

2176

QUESTION ID

S59-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 30 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q4/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q5/5

2177

QUESTION ID

S59-B1-Q5

TYPE

Short answer

CORRECT ANSWER

63 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q5/5

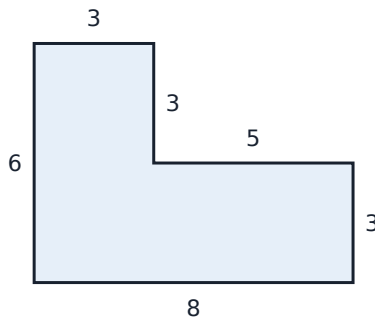
SET 59 • 6.G.A.1

BACKWARD 2

Area by splitting into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q1/5

2179

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

33 square units

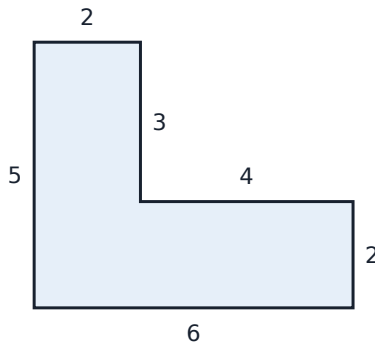
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q2/5

2180

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

18 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q2/5

True or false?

The area of an L-shaped figure can be found by adding the areas of two rectangles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q3/5

2181

QUESTION ID

S59-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q3/5

A figure is made of a 4-by-5 rectangle and a 2-by-3 rectangle that do not overlap.

What is its area?

- A.** 14 square units **B.** 20 square units
C. 26 square units **D.** 120 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q4/5

2182

QUESTION ID

S59-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 26 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q4/5

A figure is made of a 10-by-2 rectangle and a 3-by-3 square that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q5/5

2183

QUESTION ID

S59-B2-Q5

TYPE

Short answer

CORRECT ANSWER

29 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 6.G.A.1

BACKWARD 3

Classifying quadrilaterals • Grade 5 • Geometry • 5.G.B.4

True or false?

Every rectangle is a parallelogram.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q1/5

2185

QUESTION ID

S59-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q1/5

True or false?

Every parallelogram is a rectangle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q2/5

2186

QUESTION ID

S59-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q2/5

Which quadrilateral always has four right angles?

- A.** Trapezoid
- B.** Rhombus
- C.** Parallelogram
- D.** Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q3/5

2187

QUESTION ID

S59-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q3/5

How many pairs of parallel sides does a parallelogram have?

- A.** 0
- B.** 1
- C.** 2
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q4/5

2188

QUESTION ID

S59-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q4/5

True or false?

A square is a rhombus.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q5/5

2189

QUESTION ID

S59-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q5/5

SET 59 • 6.G.A.1

BACKWARD 4

Height is perpendicular to the base • Grade 4 • Geometry • 4.G.A.1

True or false?

The height of a parallelogram is perpendicular to its base.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q1/5

2191

QUESTION ID

S59-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

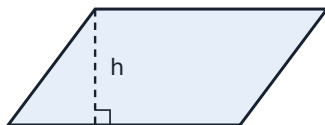
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q1/5

True or false?

The dashed segment shows the height of the parallelogram.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q2/5

2192

QUESTION ID

S59-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q2/5

Which statement describes perpendicular lines?

- A. They meet at right angles.
- B. They never meet.
- C. They meet at a 45° angle.
- D. They are always the same length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q3/5

2193

QUESTION ID

S59-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They meet at right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q3/5

True or false?

The slanted side of a parallelogram is always its height.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q4/5

2194

QUESTION ID

S59-B4-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q4/5

What is the measure of the angle between a base and its height?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q5/5

2195

QUESTION ID

S59-B4-Q5

TYPE

Short answer

CORRECT ANSWER

90 degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q5/5

SET 59 • 6.G.A.1

FORWARD 1

Area of figures made of quadrilaterals in real-world problems • Grade 7 •
Geometry • 7.G.B.6

A patio is made of a rectangle that is 10 ft by 6 ft and a trapezoid with bases of 6 ft and 4 ft and a height of 3 ft.

What is the area of the patio?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q1/5

2197

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

75 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q1/5

A wall is 12 ft by 8 ft. It has a window shaped like a parallelogram with a base of 3 ft and a height of 2 ft.

What is the area of the wall that is not window?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q2/5

2198

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

90 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q2/5

A sign is a trapezoid with bases of 1.5 m and 2.5 m and a height of 1.2 m.

What is its area?

- A.** 2.4 square meters **B.** 3.6 square meters
C. 4.8 square meters **D.** 5.2 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q3/5

2199

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.4 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q3/5

A garden is made of two parallelograms that do not overlap. Each has a base of 7 m and a height of 4 m.
What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q4/5

2200

QUESTION ID

S59-F1-Q4

TYPE

Short answer

CORRECT ANSWER

56 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q4/5

True or false?

A trapezoid with bases of 6 and 10 and a height of 5 has the same area as a rectangle that is 8 by 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q5/5

2201

QUESTION ID

S59-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q5/5

SET 59 • 6.G.A.1

FORWARD 2

Use the Pythagorean Theorem to find a missing height • Grade 8 • Geometry •
8.G.B.7

A parallelogram has a slanted side of 5 cm. The slanted side, the height, and a 3 cm piece of the base form a right triangle.

What is the height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q1/5

2203

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 cm

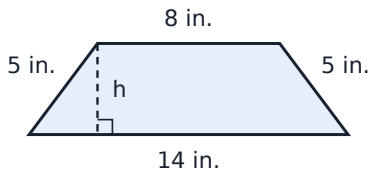
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q1/5

Find the height h of the trapezoid.

$h =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q2/5

2204

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q2/5

A ramp forms a right triangle. The ramp is 13 ft long and its base is 12 ft long.

How tall is the ramp?

- A.** 1 ft **B.** 5 ft
C. 17.7 ft **D.** 25 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q3/5

2205

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q3/5

A parallelogram has a slanted side of 10 m. The slanted side, the height, and a 6 m piece of the base form a right triangle.

What is the height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q4/5

2206

QUESTION ID

S59-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q4/5

True or false?

A trapezoid has a slanted side of 10 in. The slanted side, the height, and a 6 in. piece of the base form a right triangle. The height is 8 in.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q5/5

2207

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q5/5

Area of polygons by composing and decomposing

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectilinear figures	3.MD.C.7.d
BACKWARD 2	Missing side lengths of rectangles	4.MD.A.3
BACKWARD 3	Area of triangles	6.G.A.1
FORWARD 1	Area of composite figures in real-world problems	7.G.B.6
FORWARD 2	Volume of composite solids	8.G.C.9

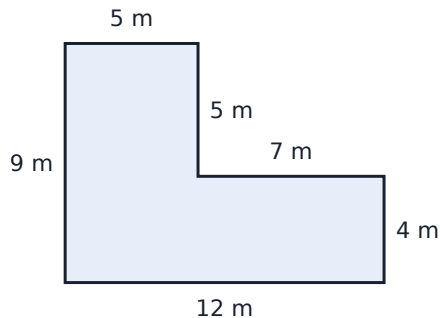
SET 60 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q1/5

2210

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

73 square meters

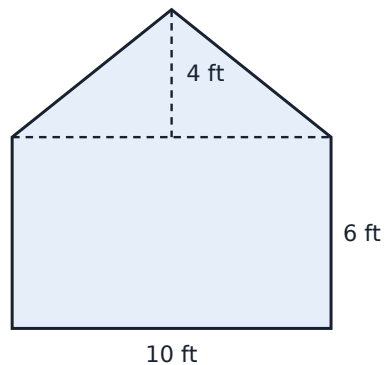
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q2/5

2211

QUESTION ID

S60-M-Q2

TYPE

Short answer

CORRECT ANSWER

80 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q2/5

A rectangular yard is 20 m by 15 m. A triangular garden in the yard has a base of 6 m and a height of 5 m.

What is the area of the yard that is NOT garden?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q3/5

2212

QUESTION ID

S60-M-Q3

TYPE

Short answer

CORRECT ANSWER

285 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q3/5

A polygon is split into a rectangle that is 8 by 3 and two triangles. Each triangle has a base of 2 and a height of 3.

What is the area of the polygon?

- A.** 24 square units **B.** 27 square units
C. 30 square units **D.** 36 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q4/5

2213

QUESTION ID

S60-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 30 square units

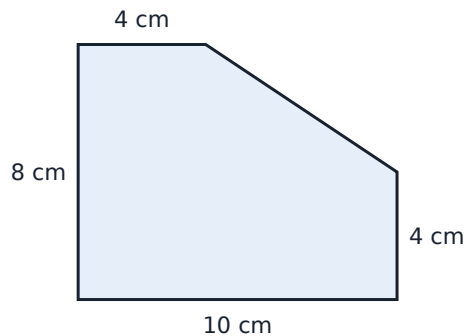
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q4/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q5/5

2214

QUESTION ID

S60-M-Q5

TYPE

Short answer

CORRECT ANSWER

68 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q5/5

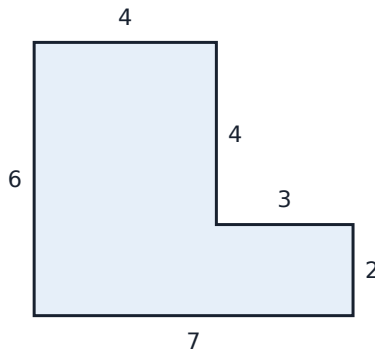
SET 60 • 6.G.A.1

BACKWARD 1

Area of rectilinear figures • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q1/5

2216

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30 square units

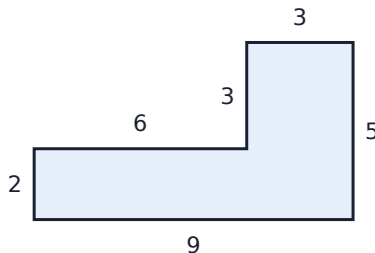
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q2/5

2217

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q2/5

True or false?

A figure made of a 5-by-2 rectangle and a 3-by-4 rectangle has an area of 22 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q3/5

2218

QUESTION ID

S60-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

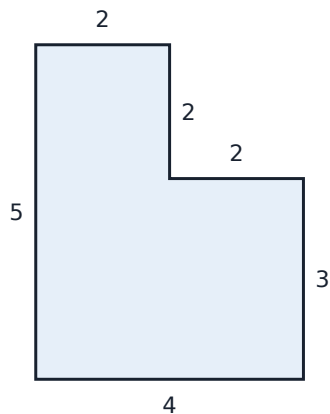
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q3/5

Which expression gives the area of the figure?

- A. 4×5
- B. $4 + 3 + 2 + 2$
- C. $4 \times 3 \times 2$
- D. $4 \times 3 + 2 \times 2$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q4/5

2219

QUESTION ID

S60-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $4 \times 3 + 2 \times 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q4/5

A figure is made of two rectangles: 6 by 3 and 2 by 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q5/5

2220

QUESTION ID

S60-B1-Q5

TYPE

Short answer

CORRECT ANSWER

26 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 6.G.A.1

BACKWARD 2

Missing side lengths of rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 48 square feet and a width of 6 feet.

What is its length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q1/5

2222

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q1/5

A rectangle has a perimeter of 30 cm and a length of 9 cm.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q2/5

2223

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q2/5

True or false?

A rectangle with an area of 36 square cm and a length of 9 cm has a width of 4 cm.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q3/5

2224

QUESTION ID

S60-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q3/5

A rectangle has an area of 56 square inches and a width of 7 inches.

What is its length?

- A.** 8 inches
- B.** 9 inches
- C.** 49 inches
- D.** 63 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q4/5

2225

QUESTION ID

S60-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8 inches

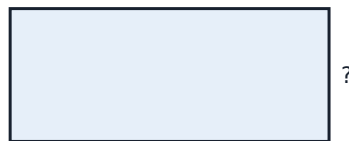
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q4/5

The rectangle has an area of 60 square meters.

What is the missing side length?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q5/5

2226

QUESTION ID

S60-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q5/5

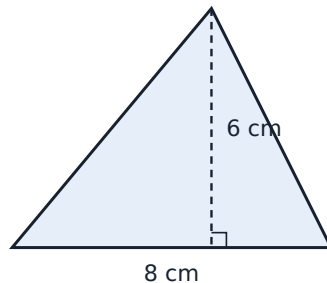
SET 60 • 6.G.A.1

BACKWARD 3

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q1/5

2228

QUESTION ID

S60-B3-Q1

TYPE

Short answer

CORRECT ANSWER

24 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q1/5

A triangle has a base of 10 m and a height of 3 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q2/5

2229

QUESTION ID

S60-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q2/5

True or false?

A right triangle with legs of 4 in. and 6 in. has an area of 24 square inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q3/5

2230

QUESTION ID

S60-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q3/5

A triangle has a base of 5 ft and a height of 8 ft.

What is its area?

- A. 13 square feet
- B. 20 square feet
- C. 26 square feet
- D. 40 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q4/5

2231

QUESTION ID

S60-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 20 square feet

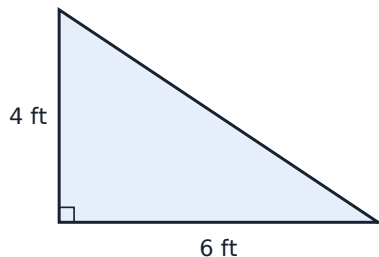
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q4/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q5/5

2232

QUESTION ID

S60-B3-Q5

TYPE

Short answer

CORRECT ANSWER

12 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • 6.G.A.1

FORWARD 1

Area of composite figures in real-world problems • Grade 7 • Geometry •
7.G.B.6

A deck is made of a rectangle that is 12 ft by 10 ft and a triangle with a base of 10 ft and a height of 6 ft.

What is the area of the deck?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q1/5

2234

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

150 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q1/5

A trapezoid-shaped garden has bases of 8 m and 14 m and a height of 5 m. It has a square pond with 2 m sides.

What is the area of the garden without the pond?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q2/5

2235

QUESTION ID

S60-F1-Q2

TYPE

Short answer

CORRECT ANSWER

51 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q2/5

A regular hexagon is split into 6 equal triangles. Each triangle has a base of 4 cm and a height of 3.5 cm.

What is the area of the hexagon?

- A.** 21 square cm **B.** 24 square cm
C. 42 square cm **D.** 84 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q3/5

2236

QUESTION ID

S60-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 42 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q3/5

An L-shaped prism has a base area of 30 square feet and a height of 4 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q4/5

2237

QUESTION ID

S60-F1-Q4

TYPE

Short answer

CORRECT ANSWER

120 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q4/5

True or false?

A figure made of a 5-by-5 square and a triangle with a base of 5 and a height of 2 has an area of 30 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q5/5

2238

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q5/5

SET 60 • 6.G.A.1

FORWARD 2

Volume of composite solids • Grade 8 • Geometry • 8.G.C.9

A silo is a cylinder with a radius of 3 m and a height of 10 m, topped by a hemisphere with a radius of 3 m.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q1/5

2240

QUESTION ID

S60-F2-Q1

TYPE

Short answer

CORRECT ANSWER

108π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q1/5

A cone has a radius of 3 in. and a height of 4 in. A hemisphere with a radius of 3 in. sits on top.

What is the total volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q2/5

2241

QUESTION ID

S60-F2-Q2

TYPE

Short answer

CORRECT ANSWER

30π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q2/5

A cone with a radius of 2 and a height of 3 is cut out of a cylinder with a radius of 2 and a height of 6.

What volume is left?

- A.** 20π **B.** 28π
C. 24π **D.** 4π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q3/5

2242

QUESTION ID

S60-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q3/5

A cylinder and a cone each have a radius of 1 and a height of 2.

How much more volume does the cylinder have, in terms of π ?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q4/5

2243

QUESTION ID

S60-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$4\pi/3$

GRADING NOTE $1 \frac{1}{3} \pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q4/5

True or false?

A sphere with a radius of 3 has the same volume as a cylinder with a radius of 3 and a height of 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q5/5

2244

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q5/5

Find volume by packing a prism with unit-fraction cubes

MAIN	Grade 6 main question	6.G.A.2
BACKWARD 1	Volume by counting unit cubes	5.MD.C.4
BACKWARD 2	The volume of one small cube	5.NF.B.4.a
BACKWARD 3	Volume as the number of unit cubes	5.MD.C.3.b
FORWARD 1	Volume of prisms in real-world problems	7.G.B.6
FORWARD 2	Volume of cylinders, cones, and spheres	8.G.C.9

SET 61 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

How many cubes with $\frac{1}{2}$ -inch edges fill a box that is 2 in. by $1\frac{1}{2}$ in. by 1 in.?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q1/5

2247

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

24 cubes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q1/5

A box is 3 in. by 2 in. by $1\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill the box?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q2/5

2248

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

72 cubes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q2/5

True or false?

A prism is packed with 48 cubes that each have $\frac{1}{2}$ -unit edges. Its volume is 6 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q3/5

2249

QUESTION ID

S61-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q3/5

A prism is packed with 64 cubes that each have $\frac{1}{4}$ -inch edges.

What is the volume of the prism?

- A.** $\frac{1}{4}$ cubic inch **B.** 1 cubic inch
C. 4 cubic inches **D.** 16 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q4/5

2250

QUESTION ID

S61-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1 cubic inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

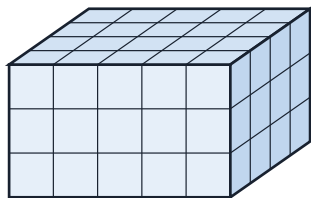
ID S61-M-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q4/5

The prism is packed with cubes that have $\frac{1}{2}$ -cm edges.

It is 5 cubes long, 4 cubes wide, and 3 cubes tall.

What is its volume?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q5/5

2251

QUESTION ID

S61-M-Q5

TYPE

Short answer

CORRECT ANSWER

7.5 cubic cm

GRADING NOTE 60 cubes $\times$ $\frac{1}{8}$ cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q5/5

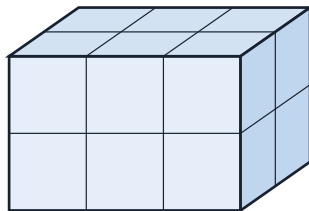
SET 61 • 6.G.A.2

BACKWARD 1

Volume by counting unit cubes • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q1/5

2253

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 cubic units

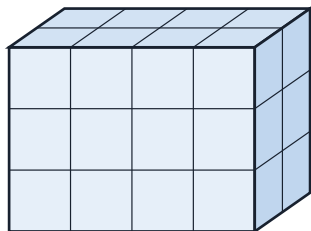
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q1/5

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q2/5

2254

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

24 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q2/5

True or false?

A prism made of 2 layers with 6 unit cubes in each layer has a volume of 12 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q3/5

2255

QUESTION ID

S61-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q3/5

A box holds 3 layers with 5 unit cubes in each layer.

What is its volume?

- A. 8 cubic units
- B. 15 cubic units
- C. 35 cubic units
- D. 53 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q4/5

2256

QUESTION ID

S61-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 15 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

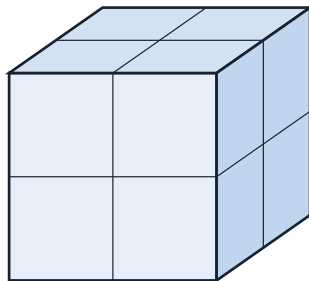
ID S61-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q4/5

Each cube is 1 cubic unit.

What is the volume of the prism?

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q5/5

2257

QUESTION ID

S61-B1-Q5

TYPE

Short answer

CORRECT ANSWER

8 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • 6.G.A.2

BACKWARD 2

The volume of one small cube • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q1/5

2259

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q1/5

Multiply.

$$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q2/5

2260

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q2/5

True or false?

$$\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q3/5

2261

QUESTION ID

S61-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q3/5

Multiply.

$$24 \times \frac{1}{8}$$

- A. $\frac{1}{3}$
- B. 3
- C. 32
- D. 192

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q4/5

2262

QUESTION ID

S61-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q4/5

Multiply.

$$64 \times \frac{1}{64}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q5/5

2263

QUESTION ID

S61-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 6.G.A.2

BACKWARD 3

Volume as the number of unit cubes • Grade 5 • Measurement & Data •
5.MD.C.3.b

True or false?

A solid that can be packed with 20 unit cubes with no gaps has a volume of 20 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q1/5

2265

QUESTION ID

S61-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q1/5

A box is filled with 36 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 6 cubic units **B.** 12 cubic units
C. 36 cubic units **D.** 36 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q2/5

2266

QUESTION ID

S61-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 36 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q2/5

A box is filled with 45 unit cubes with no gaps.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q3/5

2267

QUESTION ID

S61-B3-Q3

TYPE

Short answer

CORRECT ANSWER

45 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q3/5

True or false?

Volume is measured in square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q4/5

2268

QUESTION ID

S61-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q4/5

Which unit could be used to measure volume?

- A.** Square centimeters
- B.** Cubic centimeters
- C.** Centimeters
- D.** Kilograms

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q5/5

2269

QUESTION ID

S61-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q5/5

SET 61 • 6.G.A.2

FORWARD 1

Volume of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 5 in. and a height of 4 in. The prism is 8 in. long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q1/5

2271

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

80 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q1/5

A prism has a base area of $12\frac{1}{2}$ square cm and a height of 4 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q2/5

2272

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q2/5

A cube has edges of 2.5 cm.

What is its volume?

- A.** 6.25 cubic cm
- B.** 7.5 cubic cm
- C.** 15 cubic cm
- D.** 15.625 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q3/5

2273

QUESTION ID

S61-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15.625 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q3/5

An aquarium is 30 in. long, 12 in. wide, and 16 in. tall. It is filled with water to a height of 12 in.

What is the volume of the water?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q4/5

2274

QUESTION ID

S61-F1-Q4

TYPE

Short answer

CORRECT ANSWER

4,320 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q4/5

True or false?

A triangular prism with a base area of 15 square units and a length of 6 units has a volume of 90 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q5/5

2275

QUESTION ID

S61-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q5/5

SET 61 • 6.G.A.2

FORWARD 2

Volume of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 5 and a height of 4.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q1/5

2277

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

100π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

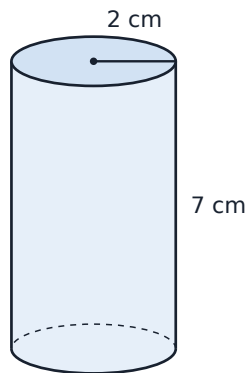
ID S61-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q1/5

Find the volume of the cylinder.

Use 3.14 for π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q2/5

ID S61-F2-Q2

2278

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

87.92 cubic cm

GRADING NOTE 28π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q2/5

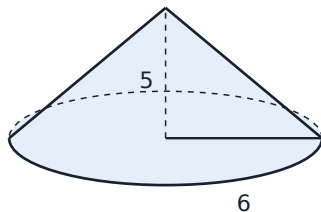
ID S61-F2-Q2

2278

A cone has a radius of 6 and a height of 5.

What is its volume?

- A.** 150π **B.** 180π
C. 30π **D.** 60π



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q3/5

2279

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q3/5

A sphere has a radius of 6.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q4/5

2280

QUESTION ID

S61-F2-Q4

TYPE

Short answer

CORRECT ANSWER

288π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q4/5

True or false?

A cylinder has a greater volume than a cone with the same radius and height.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q5/5

2281

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q5/5

Apply $V = l w h$ and $V = b h$ with fractional edge lengths

MAIN

Grade 6 main question

6.G.A.2

BACKWARD 1

Volume formula with whole numbers

5.MD.C.5.b

BACKWARD 2

Multiplying mixed numbers

5.NF.B.6

BACKWARD 3

Multiplying fractions

5.NF.B.4.a

FORWARD 1

Volume of triangular and rectangular prisms

7.G.B.6

FORWARD 2

Volume of cylinders using $V = Bh$

8.G.C.9

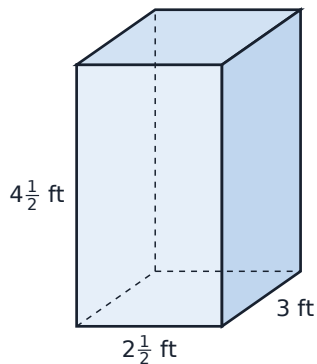
SET 62 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

Find the volume of the rectangular prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q1/5

ID S62-M-Q1

2284

QUESTION ID

S62-M-Q1

TYPE

Short answer

CORRECT ANSWER

33 $\frac{3}{4}$ cubic feet

GRADING NOTE 33.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q1/5

ID S62-M-Q1

2284

A box is $1\frac{1}{2}$ ft long, 2 ft wide, and $\frac{3}{4}$ ft tall.

What is its volume?

- A.** $2\frac{1}{4}$ cubic feet **B.** $2\frac{3}{4}$ cubic feet
C. 3 cubic feet **D.** $4\frac{1}{4}$ cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q2/5

2285

QUESTION ID

S62-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 2 1/4 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q2/5

Use $V = l \times w \times h$ to find the volume of a prism with
 $l = 5$ cm, $w = 2\frac{1}{2}$ cm, and $h = 1\frac{1}{5}$ cm.

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q3/5

2286

QUESTION ID

S62-M-Q3

TYPE

Short answer

CORRECT ANSWER

15 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q3/5

A cube has edges of $1\frac{1}{2}$ feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q4/5

2287

QUESTION ID

S62-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 $\frac{3}{8}$ cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q4/5

A prism has a base area of $4\frac{1}{2}$ square inches and a height of $2\frac{2}{3}$ inches.

Use $V = b \times h$ to find its volume.

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q5/5

2288

QUESTION ID

S62-M-Q5

TYPE

Short answer

CORRECT ANSWER

12 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q5/5

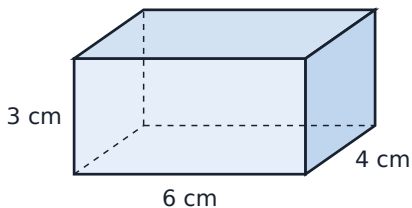
SET 62 • 6.G.A.2

BACKWARD 1

Volume formula with whole numbers • Grade 5 • Measurement & Data •
5.MD.C.5.b

Find the volume of the prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q1/5

2290

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

72 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q1/5

A prism has a length of 10 m, a width of 2 m, and a height of 5 m.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q2/5

2291

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

100 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q2/5

True or false?

A cube with 3-inch edges has a volume of 27 cubic inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q3/5

2292

QUESTION ID

S62-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q3/5

A box is 8 in. by 5 in. by 2 in.

What is its volume?

- A. 15 cubic inches
- B. 40 cubic inches
- C. 80 cubic inches
- D. 160 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q4/5

2293

QUESTION ID

S62-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 80 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q4/5

A prism has a base area of 20 square cm and a height of 6 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q5/5

2294

QUESTION ID

S62-B1-Q5

TYPE

Short answer

CORRECT ANSWER

120 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q5/5

SET 62 • 6.G.A.2

BACKWARD 2

Multiplying mixed numbers • Grade 5 • Number & Operations–Fractions •
5.NF.B.6

Multiply.

$$2\frac{1}{2} \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q1/5

2296

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q1/5

Multiply.

$$1\frac{1}{2} \times \frac{3}{4}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q2/5

2297

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$9/8 = 1\frac{1}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q2/5

True or false?

$$2\frac{1}{2} \times 2 = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q3/5

2298

QUESTION ID

S62-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q3/5

$$3 \times 4\frac{1}{2} = ?$$

A. $7\frac{1}{2}$

B. $12\frac{1}{6}$

C. $12\frac{1}{2}$

D. $13\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q4/5

2299

QUESTION ID

S62-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q4/5

Multiply.

$$1\frac{1}{3} \times 1\frac{1}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q5/5

2300

QUESTION ID

S62-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • 6.G.A.2

BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{2}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q1/5

2302

QUESTION ID

S62-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q1/5

Multiply.

$$\frac{5}{2} \times \frac{9}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q2/5

2303

QUESTION ID

S62-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$45/4 = 11 \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q2/5

True or false?

$$\frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q3/5

2304

QUESTION ID

S62-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q3/5

$$\frac{1}{2} \times \frac{4}{5} = ?$$

A. $\frac{2}{5}$

B. $\frac{4}{7}$

C. $\frac{5}{7}$

D. $\frac{8}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q4/5

2305

QUESTION ID

S62-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q4/5

Multiply.

$$\frac{6}{5} \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q5/5

2306

QUESTION ID

S62-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q5/5

SET 62 • 6.G.A.2

FORWARD 1

Volume of triangular and rectangular prisms • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 9 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q1/5

2308

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

48 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q1/5

A tent is a triangular prism. Its triangle has a base of 2.4 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q2/5

2309

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5.4 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q2/5

A prism has a base area of 7.5 square feet and a height of 2.4 feet.

What is its volume?

- A.** 9.9 cubic feet
- B.** 15 cubic feet
- C.** 18 cubic feet
- D.** 180 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q3/5

2310

QUESTION ID

S62-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 18 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q3/5

A box is 4.5 cm by 2 cm by 3.2 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q4/5

2311

QUESTION ID

S62-F1-Q4

TYPE

Short answer

CORRECT ANSWER

28.8 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q4/5

True or false?

A triangular prism with a triangle area of 10 square inches and a length of 7 inches has a volume of 70 cubic inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q5/5

2312

QUESTION ID

S62-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q5/5

SET 62 • 6.G.A.2

FORWARD 2

Volume of cylinders using $V = Bh$ • Grade 8 • Geometry • 8.G.C.9

A cylinder has a base area of 12π square cm and a height of 5 cm.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q1/5

2314

QUESTION ID

S62-F2-Q1

TYPE

Short answer

CORRECT ANSWER

60π cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q1/5

A can has a radius of 3 in. and a height of 8 in.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q2/5

2315

QUESTION ID

S62-F2-Q2

TYPE

Short answer

CORRECT ANSWER

72π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q2/5

A cylinder has a radius of 1.5 m and a height of 4 m.

What is its volume?

- A. 12π cubic meters
- B. 9π cubic meters
- C. 3π cubic meters
- D. 6π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q3/5

2316

QUESTION ID

S62-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9π cubic meters

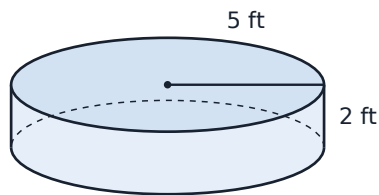
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q3/5

Find the volume of the cylinder in terms of π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q4/5

2317

QUESTION ID

S62-F2-Q4

TYPE

Short answer

CORRECT ANSWER

50π cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q4/5

True or false?

Doubling the height of a cylinder doubles its volume.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q5/5

2318

QUESTION ID

S62-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q5/5

Draw polygons in the coordinate plane from the coordinates of their vertices

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Plotting points in all four quadrants	6.NS.C.6.c
BACKWARD 2	Attributes of quadrilaterals	3.G.A.1
BACKWARD 3	Naming shapes by sides and angles	2.G.A.1
FORWARD 1	Draw shapes with given conditions	7.G.A.2
FORWARD 2	Draw images of polygons under transformations	8.G.A.3

SET 63 • 6.G.A.3

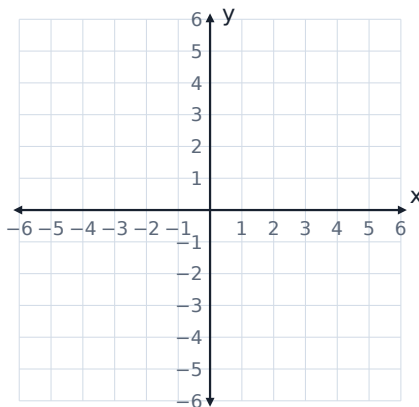
MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

Plot $A(-3, 2)$, $B(4, 2)$, $C(4, -3)$, and $D(-3, -3)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q1/5

2321

QUESTION ID

S63-M-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

GRADING NOTE Graph: $A(-3, 2)$, $B(4, 2)$, $C(4, -3)$, $D(-3, -3)$ connected in order

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

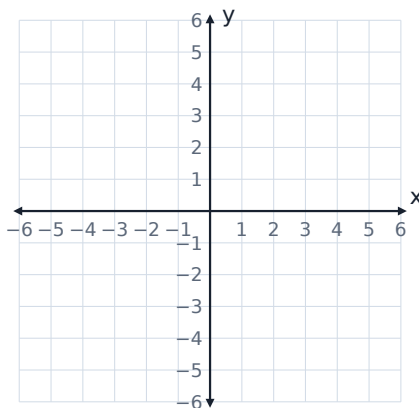
ID S63-M-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q1/5

Plot $P(0, 4)$, $Q(3, -2)$, and $R(-3, -2)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q2/5

2322

QUESTION ID

S63-M-Q2

TYPE

Short answer

CORRECT ANSWER

A triangle (isosceles)

GRADING NOTE Graph: $P(0, 4)$, $Q(3, -2)$, $R(-3, -2)$ connected

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

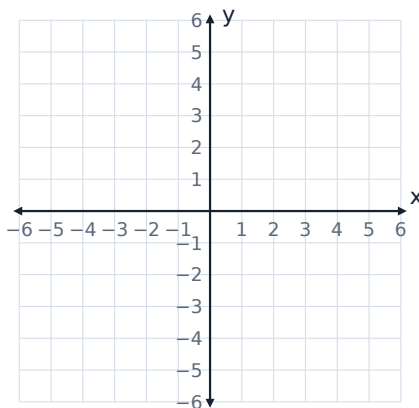
ID S63-M-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q2/5

Plot $E(-2, 1)$, $F(2, 1)$, $G(4, -3)$, and $H(-4, -3)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q3/5

2323

QUESTION ID

S63-M-Q3

TYPE

Short answer

CORRECT ANSWER

A trapezoid

GRADING NOTE Graph: $E(-2, 1)$, $F(2, 1)$, $G(4, -3)$, $H(-4, -3)$ connected; EF is parallel to HG

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

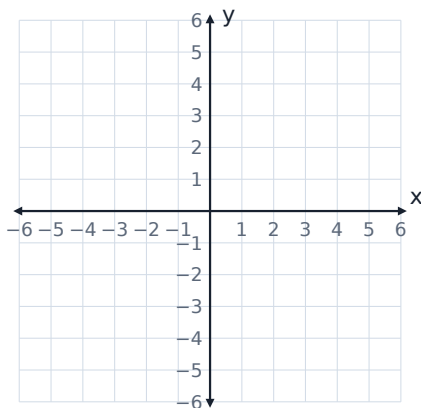
ID S63-M-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q3/5

Plot $J(-1, 3)$, $K(3, 3)$, $L(3, -1)$, and $M(-1, -1)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q4/5

2324

QUESTION ID

S63-M-Q4

TYPE

Short answer

CORRECT ANSWER

A square

GRADING NOTE Graph: $J(-1, 3)$, $K(3, 3)$, $L(3, -1)$, $M(-1, -1)$ connected; each side is 4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

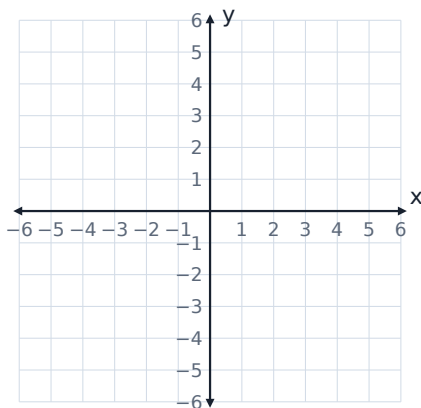
ID S63-M-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q4/5

Plot $W(-5, -2)$, $X(-2, 3)$, $Y(4, 3)$, and $Z(1, -2)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q5/5

2325

QUESTION ID

S63-M-Q5

TYPE

Short answer

CORRECT ANSWER

A parallelogram

GRADING NOTE Graph: $W(-5, -2)$, $X(-2, 3)$, $Y(4, 3)$, $Z(1, -2)$ connected; WZ and XY are parallel and 6 units long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

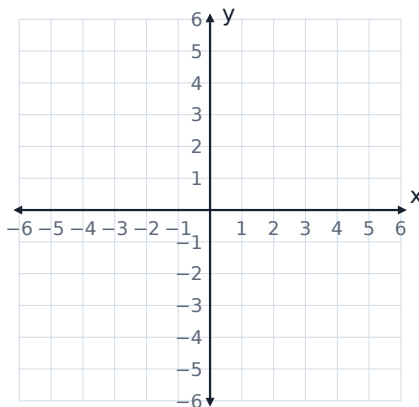
Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q5/5

SET 63 • 6.G.A.3

BACKWARD 1

Plotting points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

Plot and label point A($-3, 2$).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q1/5

2327

QUESTION ID

S63-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

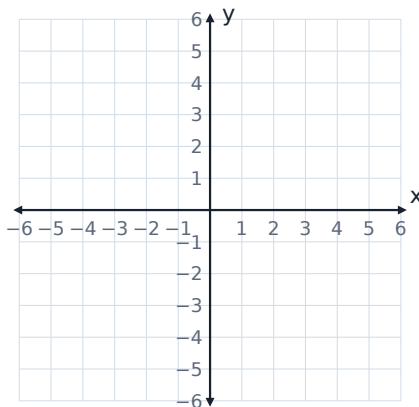
**Point A plotted at
($-3, 2$) and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q1/5

Plot and label point B(4, -3).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q2/5

2328

QUESTION ID

S63-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Point B plotted at (4, -3) and labeled

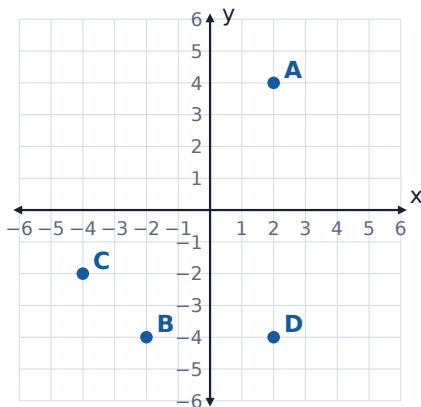
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q2/5

Which point is at $(-2, -4)$?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q3/5

2329

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

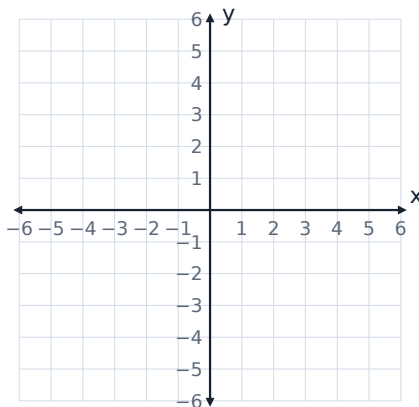
B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q3/5

Plot and label point C(0, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q4/5

2330

QUESTION ID

S63-B1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Point C plotted at (0, 4) on the y-axis and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

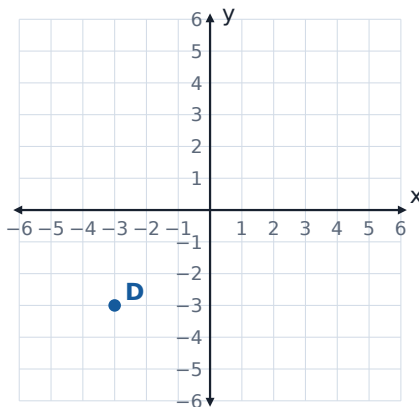
ID S63-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q4/5

True or false?

Point D is at $(-3, -3)$.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q5/5

2331

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q5/5

SET 63 • 6.G.A.3

BACKWARD 2

Attributes of quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle has 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q1/5

2333

QUESTION ID

S63-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q1/5

True or false?

Opposite sides of a rectangle have the same length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q2/5

2334

QUESTION ID

S63-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q2/5

Which shape always has 4 sides of equal length and 4 right angles?

- A.** Square
- B.** Rectangle
- C.** Trapezoid
- D.** Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q3/5

2335

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q3/5

True or false?

Every quadrilateral has 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q4/5

2336

QUESTION ID

S63-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q4/5

How many sides does a quadrilateral have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q5/5

2337

QUESTION ID

S63-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • 6.G.A.3

BACKWARD 3

Naming shapes by sides and angles • Grade 2 • Geometry • 2.G.A.1

A shape has 3 sides and 3 angles.

What is it called?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q1/5

2339

QUESTION ID

S63-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q1/5

How many sides does a pentagon have?

- A.** 4
- B.** 5
- C.** 6
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q2/5

2340

QUESTION ID

S63-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q2/5

True or false?

A hexagon has 6 sides.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q3/5

2341

QUESTION ID

S63-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q3/5

A shape has 4 sides and 4 angles.

What is it called?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q4/5

2342

QUESTION ID

S63-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A quadrilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q4/5

Which shape has the most sides?

- A.** Pentagon
- B.** Quadrilateral
- C.** Triangle
- D.** Hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q5/5

2343

QUESTION ID

S63-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

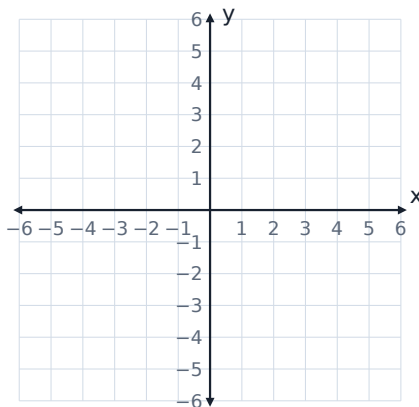
Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q5/5

SET 63 • 6.G.A.3

FORWARD 1

Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Draw a rectangle with an area of 12 square units. One vertex is at the origin.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q1/5

2345

QUESTION ID

S63-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Any rectangle with one vertex at (0, 0) and area 12, such as (0, 0), (4, 0), (4, 3), (0, 3)

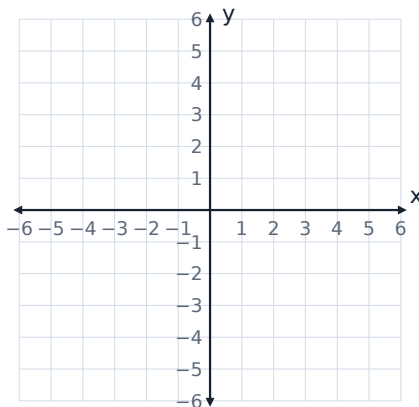
GRADING NOTE Side lengths 3×4 or 2×6 in any quadrant are correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q1/5

Draw a right triangle with legs of 4 units and 3 units. One vertex is at $(-2, -2)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q2/5

2346

QUESTION ID

S63-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A right triangle with legs 4 and 3 meeting at a right angle, with one vertex at $(-2, -2)$, such as $(-2, -2)$, $(2, -2)$, $(-2, 1)$

GRADING NOTE Any orientation with legs of 4 and 3 units is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q2/5

Can a triangle have side lengths of 2 cm, 3 cm, and 6 cm? Write yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q3/5

2347

QUESTION ID

S63-F1-Q3

TYPE

Short answer

CORRECT ANSWER

No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q3/5

A triangle has angles of 90° and 45° .

What is the third angle?

- A. 45°
- B. 55°
- C. 90°
- D. 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q4/5

2348

QUESTION ID

S63-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

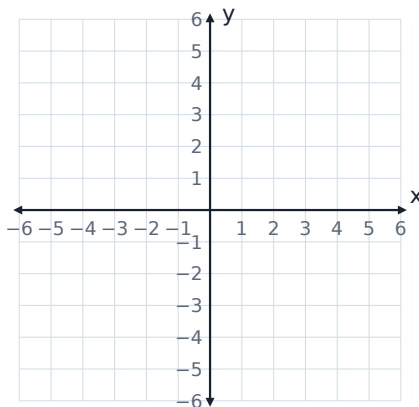
A. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q4/5

Draw a square with a side length of 5 units. One vertex is at $(-3, 1)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q5/5

2349

QUESTION ID

S63-F1-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A square with 5-unit sides and one vertex at $(-3, 1)$, such as $(-3, 1)$, $(2, 1)$, $(2, 6)$, $(-3, 6)$

GRADING NOTE Any square with side 5 using $(-3, 1)$ as a vertex that fits on the grid is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q5/5

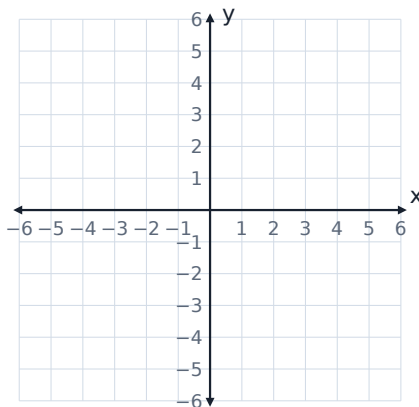
SET 63 • 6.G.A.3

FORWARD 2

Draw images of polygons under transformations • Grade 8 • Geometry •
8.G.A.3

Triangle ABC has vertices
 $A(1, 1)$, $B(4, 1)$, and $C(1, 3)$.

Draw its image after a
translation 5 units left and 2
units down.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q1/5

2351

QUESTION ID

S63-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Image triangle with
vertices $A'(-4, -1)$,
 $B'(-1, -1)$, $C'(-4, 1)$**

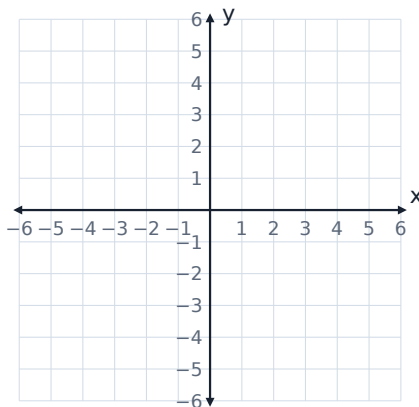
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q1/5

Rectangle PQRS has vertices $P(1, 2)$, $Q(4, 2)$, $R(4, 4)$, and $S(1, 4)$.

Draw its image after a reflection across the x-axis.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q2/5

2352

QUESTION ID

S63-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Image rectangle
with vertices $P'(1, -2)$, $Q'(4, -2)$, $R'(4, -4)$, $S'(1, -4)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q2/5

Triangle ABC has vertices A(2, 1), B(4, 3), and C(2, 3). It is rotated 180° about the origin.

What are the coordinates of B'?

B' = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q3/5

2353

QUESTION ID

S63-F2-Q3

TYPE

Short answer

CORRECT ANSWER

(-4, -3)

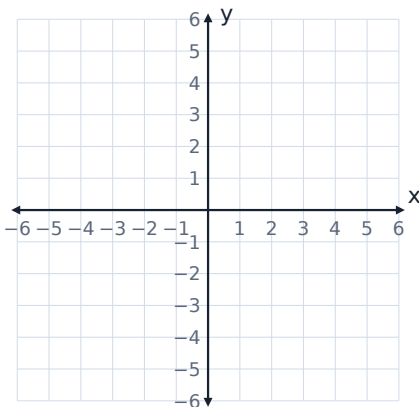
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q3/5

Square EFGH has vertices E(1, 1), F(2, 1), G(2, 2), and H(1, 2).

Draw its image after a dilation by a scale factor of 3 centered at the origin.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q4/5

2354

QUESTION ID

S63-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Image square with vertices E'(3, 3), F'(6, 3), G'(6, 6), H'(3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q4/5

True or false?

A rectangle with vertices $(1, 1)$, $(3, 1)$, $(3, 2)$, $(1, 2)$ is reflected across the y -axis. One image vertex is $(-3, 2)$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q5/5

2355

QUESTION ID

S63-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q5/5

Use coordinates to find side lengths and perimeters of polygons

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Distance between points with the same x or y	6.NS.C.8
BACKWARD 2	Absolute value	6.NS.C.7.c
BACKWARD 3	Perimeter of polygons	3.MD.D.8
FORWARD 1	Lengths between points with rational coordinates	7.NS.A.1.c
FORWARD 2	Find slanted side lengths with the Pythagorean Theorem	8.G.B.8

SET 64 • 6.G.A.3

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

A rectangle has vertices at $(-3, 2)$, $(4, 2)$, $(4, -5)$, and $(-3, -5)$.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q1/5

2358

QUESTION ID

S64-M-Q1

TYPE

Short answer

CORRECT ANSWER

28 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q1/5

One side of a polygon goes from $(-2, -3)$ to $(-2, 5)$.

How long is this side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q2/5

2359

QUESTION ID

S64-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 units

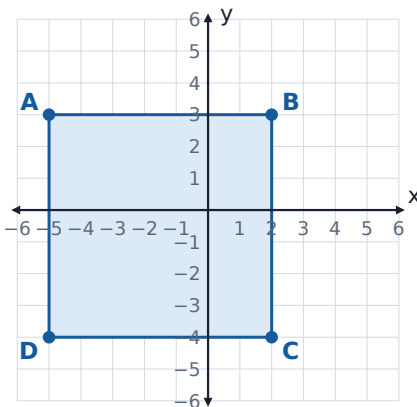
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q2/5

What is the length of side AB?

AB = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q3/5

2360

QUESTION ID

S64-M-Q3

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q3/5

A triangle has vertices at $(-4, 1)$, $(2, 1)$, and $(2, -7)$.

How long is the horizontal side?

- A. 2 units
- B. 6 units
- C. 8 units
- D. 10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q4/5

2361

QUESTION ID

S64-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q4/5

A rectangle has vertices at $(-5, -2)$, $(1, -2)$, $(1, 3)$, and $(-5, 3)$.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q5/5

2362

QUESTION ID

S64-M-Q5

TYPE

Short answer

CORRECT ANSWER

22 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q5/5

SET 64 • 6.G.A.3

BACKWARD 1

Distance between points with the same x or y • Grade 6 • The Number System
• 6.NS.C.8

Find the distance between $(-3, 2)$ and $(4, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q1/5

2364

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q1/5

Find the distance between $(1, -4)$ and $(1, 6)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q2/5

2365

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q2/5

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q3/5

2366

QUESTION ID

S64-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q3/5

What is the distance between $(2, -3)$ and $(2, -8)$?

- A.** -11
- B.** -5
- C.** 5
- D.** 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q4/5

2367

QUESTION ID

S64-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q4/5

Find the distance between $(-6, -1)$ and $(-2, -1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q5/5

2368

QUESTION ID

S64-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • 6.G.A.3

BACKWARD 2

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-5| + |2|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q1/5

2370

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q1/5

True or false?

$$|-7| = 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q2/5

2371

QUESTION ID

S64-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q2/5

$$|-3| + |4| = ?$$

- A.** -7
- B.** -1
- C.** 1
- D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q3/5

2372

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q3/5

Find the value.

$$|-6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q4/5

2373

QUESTION ID

S64-B2-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q4/5

True or false?

$$|-2| + |-2| = 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q5/5

2374

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • 6.G.A.3

BACKWARD 3

Perimeter of polygons • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of the rectangle.

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q1/5

2376

QUESTION ID

S64-B3-Q1

TYPE

Short answer

CORRECT ANSWER

20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q1/5

A square has sides of 9 inches.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q2/5

2377

QUESTION ID

S64-B3-Q2

TYPE

Short answer

CORRECT ANSWER

36 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q2/5

True or false?

A rectangle that is 10 m by 4 m has a perimeter of 28 m.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q3/5

2378

QUESTION ID

S64-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q3/5

A triangle has sides of 5 cm, 6 cm, and 7 cm.

What is its perimeter?

- A. 11 cm
- B. 18 cm
- C. 30 cm
- D. 210 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q4/5

2379

QUESTION ID

S64-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 18 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q4/5

A rectangle is 12 ft long and 5 ft wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q5/5

2380

QUESTION ID

S64-B3-Q5

TYPE

Short answer

CORRECT ANSWER

34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q5/5

SET 64 • 6.G.A.3

FORWARD 1

Lengths between points with rational coordinates • Grade 7 • The Number System • 7.NS.A.1.c

A side of a polygon goes from $(-2.5, 3)$ to $(4, 3)$.

How long is the side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q1/5

2382

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6.5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q1/5

A rectangle has vertices at $(-1.5, 2)$, $(3.5, 2)$, $(3.5, -1)$, and $(-1.5, -1)$.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q2/5

2383

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

16 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q2/5

How long is the segment from $(0, -3\frac{1}{2})$ to $(0, 1\frac{1}{4})$?

A. $-4\frac{3}{4}$

B. $2\frac{1}{4}$

C. $4\frac{3}{4}$

D. $5\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q3/5

2384

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q3/5

A square has vertices at $(-2.5, -2.5)$, $(2.5, -2.5)$, $(2.5, 2.5)$, and $(-2.5, 2.5)$.

What is the length of one side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q4/5

2385

QUESTION ID

S64-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q4/5

True or false?

The segment from $(-4.2, 1)$ to $(1.8, 1)$ is 6 units long.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q5/5

2386

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q5/5

SET 64 • 6.G.A.3

FORWARD 2

Find slanted side lengths with the Pythagorean Theorem • Grade 8 • Geometry
• 8.G.B.8

A triangle has vertices at $(0, 0)$, $(6, 0)$, and $(6, 8)$.

How long is the slanted side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q1/5

2388

QUESTION ID

S64-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 units

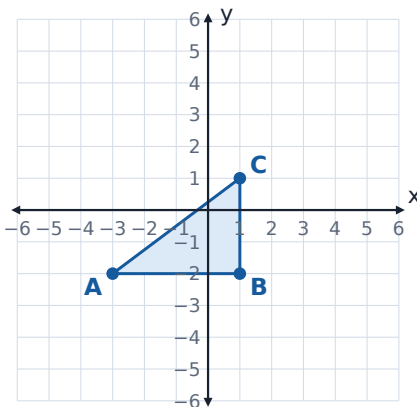
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q1/5

Find the length of side AC.

AC = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q2/5

2389

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q2/5

A parallelogram has a side from (1, 1) to (4, 5).

How long is that side?

- A. 3
- B. 5
- C. 7
- D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q3/5

2390

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q3/5

Find the perimeter of the triangle with vertices (0, 0), (5, 0), and (0, 12).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q4/5

2391

QUESTION ID

S64-F2-Q4

TYPE

Short answer

CORRECT ANSWER

30 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q4/5

True or false?

The side from $(-2, 3)$ to $(4, -5)$ is 10 units long.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q5/5

2392

QUESTION ID

S64-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q5/5

Represent three-dimensional figures using nets

MAIN	Grade 6 main question	6.G.A.4
BACKWARD 1	Faces of three-dimensional shapes	2.G.A.1
BACKWARD 2	Shapes of faces	3.G.A.1
FORWARD 1	Cross sections of three-dimensional figures	7.G.A.3
FORWARD 2	Volume of a cylinder described by its net	8.G.C.9

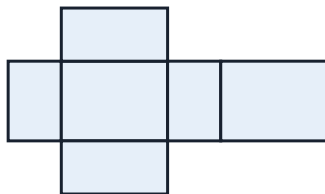
SET 65 • 6.G.A.4

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

What three-dimensional figure does the net form?

- A. Triangular prism
- B. Rectangular prism
- C. Square pyramid
- D. Cube



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q1/5

2395

QUESTION ID

S65-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Rectangular prism

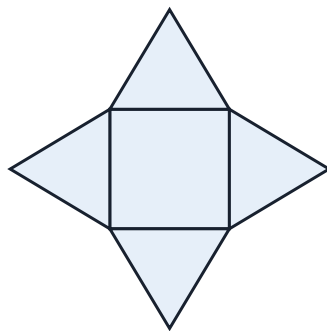
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q1/5

What three-dimensional figure does the net form?

- A. Cube
- B. Triangular pyramid
- C. Square pyramid
- D. Triangular prism



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q2/5

2396

QUESTION ID

S65-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Square pyramid

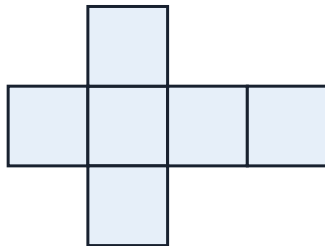
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q2/5

What three-dimensional figure does the net form?

- A. Rectangular prism
- B. Triangular prism
- C. Cube
- D. Square pyramid



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q3/5

2397

QUESTION ID

S65-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Cube

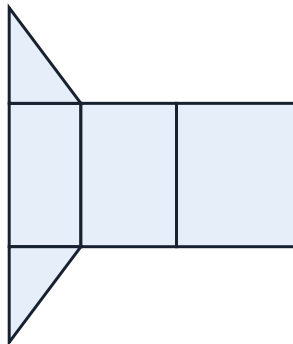
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q3/5

What three-dimensional figure does the net form?

- A. Triangular prism
- B. Triangular pyramid
- C. Square pyramid
- D. Rectangular prism



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q4/5

2398

QUESTION ID

S65-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Triangular prism

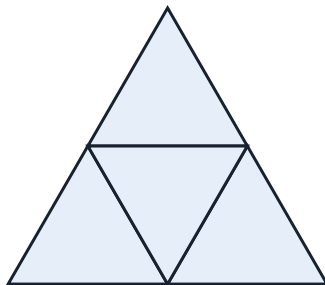
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q4/5

What three-dimensional figure does the net form?

- A. Cube
- B. Square pyramid
- C. Triangular pyramid
- D. Triangular prism



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q5/5

2399

QUESTION ID

S65-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

**C. Triangular
pyramid**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q5/5

SET 65 • 6.G.A.4

BACKWARD 1

Faces of three-dimensional shapes • Grade 2 • Geometry • 2.G.A.1